Prague University of Economics and Business

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DESIGN OF SSBI SOLUTION FOR SUPPLIER PERFORMANCE ASSESSMENT

BACHELOR THESIS

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Declaration

I hereby declare that this bachelor thesis titled "Design of SSBI solution for supplier performance assessment" is my own work. All sources and materials used have been properly acknowledged and cited. This work has not been published or submitted elsewhere for the requirement of a degree programme.

Prague, December 1, 2023 ....

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Abstract

This bachelor thesis details the design of a Self-Service Business Intelligence (SSBI) solution for assessing the performance of IT service suppliers in a corporate setting. The primary aim is to automate reporting through a Power BI dashboard, which enhances visibility into supplier performance, focusing on adherence to Underpinning Contracts (UCs). The research begins with a thorough literature review encompassing business intelligence, SSBI, and data visualization, followed by a detailed analysis of the company's current performance assessment methods and the role of underpinning contracts. The development of the dashboard involved close collaboration with company stakeholders and suppliers, prioritizing data accuracy, functionality, and ease of use. The Power BI dashboard was carefully designed to provide a detailed analysis of supplier performance, particularly in relation to incident resolution. Having undergone extensive testing, the dashboard is now actively utilized by key users, offering up-to-date insights, and enhancing decision-making processes regarding supplier management.

Keywords

SSBI, supplier performance, dashboard, Underpinning Contract, Power BI

Abstrakt

Tato bakalářská práce se zavývá návrhem SSBI (Self-Service Business Intelligence) řešení pro hodnocení výkonnosti dodavatelů IT služeb v podnikovém prostředí. Hlavním cílem je automatizace reportingu prostřednictvím dashboardu Power BI, který zvyšuje přehled o výkonnosti dodavatelů se zaměřením na dodržování Podpůrných smluv (UC). Výzkum začíná důkladným přehledem literatury zahrnujícím business intelligence, SSBI a vizualizaci dat, po němž následuje podrobná analýza současných metod hodnocení výkonnosti společnosti a role podpůrných smluv. Vývoj dashboardu zahrnoval úzkou spolupráci se zaměstnanci společnosti a dodavateli, přičemž prioritou byla přesnost údajů, funkčnost a snadnost použití. Power BI dashboard byl pečlivě navržen tak, aby poskytoval podrobnou analýzu výkonnosti dodavatelů, zejména ve vztahu k řešení incidentů. Po rozsáhlém testování je nyní dashboard aktivně využíván klíčovými uživateli společnosti, kterým nabízí aktuální informace a zlepšuje rozhodovací procesy týkající se řízení dodavatelů.

Klíčová slova

SSBI, výkon dodavatele, dashboard, Podpůrná Smlouva, Power BI

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Introduction

In the rapidly changing world of information technology (IT), relying on external suppliers for specialized services has become a widespread practice. As organizations navigate these shifts, the significance of effectively managing relationships with IT service suppliers grows increasingly evident. Central to this management strategy is the assessment of supplier performance. This not only serves as an indicator of service quality but also offers critical insights into cost efficiencies, risk profiles, and the overall vitality of the partnership between the organization and its suppliers.

This thesis examines IT service suppliers' performance within a company using the tools of Self-Service Business Intelligence (SSBI). It details the design and implementation of an SSBI solution, providing both the company's stakeholders and its suppliers with clear visibility into performance metrics and enabling efficient reporting.

Problem Statement

There is a need for decision-makers within the company to accurately assess supplier performance, particularly in terms of adherence to key performance indicators and Service Level Agreements (SLAs). Non-compliance with these metrics can have significant financial implications, making it crucial for the organization to have clear insights into areas for improvement and outstanding performance.

Objective

The primary objective of this thesis is to design and implement a Self-Service Business Intelligence (SSBI) solution that visualizes and analyses the performance of IT service suppliers within the company. From this main objective, two key sub-goals emerge.

1. Automation of Reporting: Automate the supplier performance reporting process. This automation ensures that the supplier management team does not solely rely on externally provided data. Instead, they can trust and utilize internally generated reports that are automatically produced.
2. Enhanced Visibility: Improve visibility into supplier performance for the supplier management team, internal stakeholders, and the suppliers themselves. The objective is to tailor this enhancement based on specific requirements gathered from the involved parties and to refine it using feedback from solution testing.

The objectives of this work will be considered met upon achieving specific, tangible milestones. These include the elimination of the need for manual report generation by suppliers, as all necessary data will be automatically integrated and presented through the SSBI solution. Moreover, the fulfilment of these objectives will be further validated during the feedback phase, where key users evaluate and confirm that all high and medium priority requirements have been effectively met and addressed.

Methodology

To achieve the objectives of this thesis, an agile-centric methodology was embraced, firmly rooted in iterative development cycles and continuous feedback loops. Throughout the entire process of the SSBI solution development, open communication channels were maintained with the company stakeholders, ensuring that their needs guided the development of the BI application.

The theoretical foundation of this thesis was constructed through an extensive review of academic articles, relevant books, and publications on Business Intelligence, SSBI, data visualizations, and other related topics. This foundation aims to equip the reader with a thorough understanding of the nuances of the SSBI solution and the methodology applied in the practical segment of the thesis. The SSBI solution was developed based on insights from the academic literature review, and its design was further influenced by the author's own academic studies and practical work experiences. Moreover, analysis was undertaken covering both the company's environmental and IT infrastructure.

To achieve the first subobjective, the initial step was to assess the current state of supplier performance evaluation and reporting within the company. This involved frequent meetings and discussions with potential primary users of the BI application, specifically the internal supplier management team and the suppliers themselves. Insights from these meetings were then analysed to determine the application's requirements. To address the second subobjective, access to company’s database was necessary to evaluate data availability for supplier performance assessment and to facilitate the design of the BI application. Furthermore, a preliminary data model was developed to structure the data appropriately.

Following these steps, a prototype of the dashboard was constructed. Throughout the development process, particular emphasis was placed on evaluating user friendliness, functionality, data accuracy, and the dashboard's filtering capabilities. Regular iterations and updates were based on user feedback to ensure alignment with the company's evolving needs. In addition to the communication about the BI application's requirements, the dashboard underwent testing by both internal company users and suppliers. The objective of this testing was to evaluate the solution's effectiveness and ensure it aligned with the key users' predefined requirements. Feedback sessions also highlighted potential areas for enhancements.

Assumptions and Limitations

A firm understanding of theory on how to implement a Self-Service Business Intelligence (SSBI) solution within a business setting is required. On the practical side, it necessitates proficiency in data modeling and transformation, with a focus on the Power Query editor and the DAX language. Proficiency in MS Power BI Desktop and Power BI Service is essential for dashboard creation and its subsequent publication and distribution. In a corporate context, an awareness of data security and governance is vital, particularly the implementation of Row-Level Security (RLS). Another prerequisite is the access to the company's internal ticketing tool, database, and other information resources provided by the company. It is also crucial to ensure a collaborative relationship with the key users of the dashboard.

The primary constraint of this thesis is that the design of the SSBI solution was restricted to technologies that were already approved, available, and licensed within the company. A significant limitation is the requirement for anonymity as the company's identity must remain confidential. Additionally, the author doesn’t have the right to publish or disclose any internal data. To adhere to these requirements and protect sensitive information, all data used in the development of the dashboard, such as supplier names, numbers, and other proprietary details, has been modified or anonymized.

A further constraint was the limited time allocated for this project. This time limitation was partly due to the necessity of adhering to internal regulations and established processes within the company. These procedures, including obtaining necessary approvals and access requests, contributed to the time-consuming nature of this thesis. Consequently, considering the vast number of metrics available for evaluating IT service suppliers' performance within the company, this thesis focused solely on two specific metrics across various services: the resolution and response times of incidents, differentiated by their priorities. Furthermore, the data is limited to the year 2023.

Outline

This thesis is structured with an introduction, two core chapters, and a conclusion. The introduction defines the primary objective of the work, its sub-goals, and highlights the assumptions, limitations, expected outcomes, and potential contributions of the study.

The central chapters encompass both theoretical and practical perspectives. The theoretical segment comprises a literature review and an exposition on business intelligence, equipping the reader with foundational knowledge in this field. On the other hand, the practical segment delves into the introductory study giving an overview of the company and its supplier performance assessment methods, dashboard design and implementation, and a discussion on the research findings.

The conclusion synthesizes the study's significance, its key findings, and implications for the company's decision-making framework. It further evaluates if the thesis achieved the goals set forth in the introduction.

Anticipated Outcomes and Potential Contributions

The main outcome of this thesis is a Power BI dashboard designed to be used on a daily basis by both internal employees and suppliers. This tool allows better analysis of supplier performance and offers enhanced visibility into potential areas for improvement and data-driven insights for decision-making. At its core, the dashboard automates the reporting process for various service levels provided by suppliers which eliminates the reliance on manually generated reports.

Further contributions aim to equip the reader with theoretical knowledge on the topics of business intelligence and data visualization techniques, provide insights into the current state of reporting and supplier performance monitoring within the company, and offer an overview of user requirements. An additional contribution to the company is the creation of technical documentation, which acts as a structured guide for future modifications and deployments of the dashboard.

1 Literature Review

This chapter provides a comprehensive literature review sourced from various academic materials that lay the foundation for this bachelor thesis. The resources consulted range from books and different academic articles to master's theses, all focusing on related topics. The selection of these sources was guided by the keywords specified in the abstract, including Business Intelligence, Power BI, dashboard, data visualization, SSBI (Self-Service Business Intelligence), supplier performance, and UC (Underpinning Contract).

1.1 Academic Literature

Self Service Business Intelligence: jak si vytvořit vlastní analytické, plánovací a reportingové aplikace (Pour et al., 2018)

This book offers real-world examples of SSBI applications in the domains of business management and planning. Not only does it cover SSBI tools and their respective use cases, but it also imparts theoretical understanding on topics like data modeling, the processes of designing and architecting data warehouses, and data visualization techniques. Importantly, the book details the design of an SSBI solution in Power BI, the tool chosen for the practical segment of this thesis. This book has been instrumental in laying the methodological groundwork for the SSBI solution.

Business analytika v praxi (Potančok et al., 2020)

"Business analytika v praxi" by Martin Potančok, Jan Pour, and Veronika Chramostová (2020), delves into the growing importance of business analytics in both corporate management and information technology. It addresses the common analytical challenges faced by modern businesses and offers various solutions to overcome these obstacles. The book extensively discusses the business landscape and how companies can use business analytics to meet their specific needs. It also explores how company strategies are connected to Business Intelligence, providing a solid understanding of this relationship. Information from this book was used as the theoretical foundation of the thesis.

Introducing Microsoft Power BI (Ferarri, Russo, 2016)

The book lays down solid practical knowledge of Microsoft Power BI. The insights helped to understand how data visualization and analytics tools like Power BI can be employed to improve business processes, aid in decision-making, and drive business success. The book's easy-to-follow, step-by-step guide approach towards explaining the functionalities of Power BI makes it a reliable reference for exploring the application of this tool in a real-world business environment. Information from this book was used to create the Power BI dashboard which is described in the practical part of this thesis.

The Complete Guide to IT Service Level Agreements: Aligning IT Services to Business Needs (Hiles et al., 2016)

In the context of the thesis, this book provides a rich framework for understanding the importance and the process of aligning IT services with business needs through well-structured Service Level Agreements. It serves as a significant reference for discussing the theoretical and practical aspects of designing and implementing IT SLAs, thereby ensuring that the IT services are in sync with the organizational objectives and customer expectations. This alignment, as discussed in the book, is crucial for preventing disputes, justifying investments, and ultimately, achieving business success. The knowledge gained from this book is utilized in the introductory study of the thesis and in the construction of the SSBI solution.

ITIL Foundation: ITIL 4 edition (ITIL, 2019)

The book "ITIL Foundation: ITIL 4 edition" (2019) provides valuable insights into managing incidents and maintaining Service Level Agreements (SLAs) in IT Service Management. It outlines key strategies for handling incidents and ensuring service levels meet business needs. In the thesis, this book served as a helpful resource for understanding incident management and SLAs. Its guidance was crucial in discussing these topics in the introductory study and in developing the SSBI solution, ensuring it adheres to recognized ITSM practices.

1.2 University Theses Research

Design of internal reporting and its implementation in the company (Mičáková, 2019)

Zuzana Mičáková (2023), in her master thesis titled "Design of internal reporting and its implementation in the company" designs and implements a new reporting solution for a company, with the objective of automating processes within selected departments. Following a thorough analysis, the author crafted a logical design for the solution, which subsequently formed the foundation for the dashboard. The author underscores the efficiency of data parametrization and the importance of generating technical documentation to facilitate future enhancements to the solution.

Development and implementation of an internal dashboard for an IT team in a pharmaceutical company (Sharapova, 2022)

In her master thesis titled "Development and implementation of an internal dashboard for an IT team in a pharmaceutical company", Diana Sharapova (2022) explores the creation and deployment of an internal dashboard for a pharmaceutical firm's IT team using Power BI. The thesis outputs a functional dashboard aimed at enhancing reporting within the team. Through an initial analysis, the author outlines the current reporting scenario, investigates other internal dashboards within the company, and evaluates the newly designed dashboard to identify areas for improvement. Emphasis is placed on the architectural design of the solution, user requirements analysis, and evaluation of the final dashboard to foster future improvements in reporting within the company.

1.3 Other Sources

It is important to highlight that a significant contribution to this bachelor thesis came from the meetings and discussions conducted with the company employees. Gaining an understanding of a company's inner workings is crucial before developing any applications that may be utilized therein. To this end, requirements were gathered directly with the supplier management team. However, discussions did not end at this stage; subsequent feedback was collected from various company suppliers and internal stakeholders regarding the dashboard, its further enhancements, and maintenance. This iterative feedback mechanism not only enriched the development process but also rooted the study in practical insights and real-world applicability. Additionally, the incorporation of information from various internet articles and sources further augmented the research, providing a broader perspective and aligning the study with current trends and practices.

2 Business Intelligence

Business Intelligence (BI) is a way of utilizing information technologies for analytical, planning, and decision-making processes within a company. It encompasses a collection of processes, knowledge, applications, and technologies designed to enhance a company's operational activities effectively and efficiently. BI aids processes across all organizational levels and areas of business management, including financial management, controlling, sales, purchasing, marketing, human resource management, and others (Pour et al., 2018).

The foundational era of business intelligence began in 1958 when IBM researcher Hans Peter Luhn first introduced the term "business intelligence." It was characterized by a basic understanding and utilization of data for business advantage. Over the years, the definition and scope of BI have significantly evolved, embracing modern technology and data analytics techniques (Luhn, 1958).

The first generation of BI largely revolved around the IT department, which served as the custodian of enterprise data. The process involved Extract, Transform, Load (ETL) operations to aggregate data into databases or data warehouses for analytics and reporting. However, this process was quite rigid and could take a substantial amount of time, often days or even weeks. This was due to dependencies on skilled IT personnel for query execution and report generation (Yellowfin, 2022).

Modern BI has expanded to cover a broader spectrum of operations. It now encompasses processes and methods of collecting, storing, and analysing business data to optimize performance. In the face of rapidly growing data volumes, especially with Big Data and IoT, the field of BI has further evolved towards augmented analytics. This new approach uses artificial intelligence and machine learning to enable real-time analytics and data visualization techniques (Lees, 2021).

According to Turban et al. (2011), *"Business Intelligence is an umbrella term that combines architectures, tools, databases, analytical tools, applications, and methodologies."* This definition underscores the comprehensive nature of BI, encapsulating both the tools and practices involved.

Another perspective is presented by Howson (2007), who states that *"Business Intelligence is a set of theories, methodologies, architectures, and technologies that transform raw data into meaningful and useful information for business purposes."*

BI plays a key role in optimizing organizational operations. Companies can utilize it to enhance decision-making: reveal emerging opportunities, expose potential risks, provide industry insights, and strengthen decision-making frameworks (Niu et al., 2021).

2.1 Architecture

Novotný, Pour, and Slánský in their book "Business Intelligence: Jak využít bohatství ve vašich datech" (2005) showcase a general concept of BI architecture solution (see Figure 1.). The figure depicts a schematic representation of BI components and their interrelationships, including data sources, analytical tools, and user interfaces.

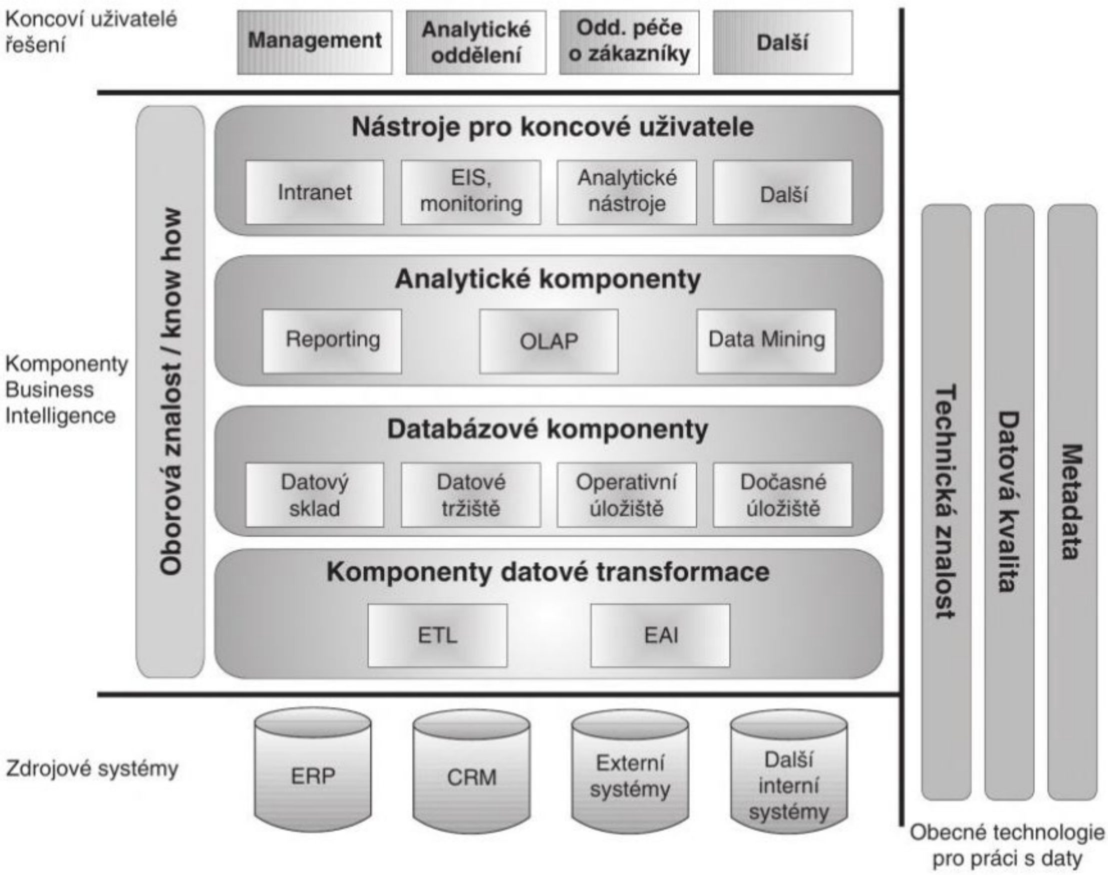


Figure 1: General concept of BI architecture. Reprinted from "Business Intelligence: Jak využít bohatství ve vašich datech" by Novotný et al., 2005.

2.2 Components

The components of BI have evolved over time, with certain key elements forming the core of most BI solutions. These include data warehouses, data marts, data pumps (ETL/ELT), OLAP solutions, analytical applications, reporting, data mining, information management (i.e., data quality, metadata, security), etc. (Pour et al., 2018). Below is a further elaboration on each component.

Data Warehouse

Data warehouses are specialized systems designed primarily for analytical processing. Unlike standard databases, which primarily handle transactional operations, data warehouses consolidate data from diverse sources into a single, unified structure. This centralization supports large-scale queries and complex analysis. Data within warehouses is often structured using a dimensional or star schema, which allows for easier querying and reporting (Inmon, 2005).

Data Mart

Data marts can be understood as subsets of a data warehouse, focusing on a specific business area, such as finance or marketing. These were created to address the specific analytical needs of certain departments. They can source data either from the main data warehouse or directly from transactional systems. The main advantage of a data mart is its ability to provide faster and more focused insights for its specific domain (Kimball & Ross, 2013).

Data Lake

A data lake is a repository that maintains an extensive volume of unprocessed data in its original format for future use. It is designed to handle high volumes of data that are too varied and fast-changing for traditional data management systems. The concept allows for the storage of heterogeneous data, making it a flexible option for big data analytics, particularly in academic settings where varied and voluminous datasets are common (Anderson, 2022).

OLAP

Online Analytical Processing (OLAP) facilitates multidimensional data analysis, enabling a more nuanced examination compared to traditional two-dimensional table views. Through its cube structure, OLAP organizes data for swift querying, offering a platform to analyse sales data by various dimensions like time, region, and product concurrently. This technology enhances decision-making by providing a multidimensional perspective, which is crucial in contemporary data-driven business environments (Codd, Codd & Salley, 1993).

OLAP technology is defined in several modern resources. For instance, a chapter in a book titled "Data Mining and Data Warehousing" explains the essence of OLAP, distinguishing it from Online Transaction Processing (OLTP) and data mining. The chapter emphasizes the representation of a multidimensional view of data, which is pivotal in real-world applications (Bhatia, 2019). Another resource, "On-line Analytical Processing Systems for Business," underscores OLAP's role as a novel approach in information system technology that augments decision-making processes. The book highlights multidimensional analysis as a fundamental feature of OLAP, which exceeds conventional two-dimensional analysis, thus providing users with rapid querying capabilities (Thierauf, 2001).

Data Pumps (ETL/ELT)

Data Pumps (ETL/ELT): The act of transporting data from operational systems to analytical systems is rarely straightforward. ETL (Extract, Transform, Load) and ELT (Extract, Load, Transform) are methodologies to facilitate this process. ETL involves extracting data from source systems, then transforming it to ensure uniformity, and loading the standardized data into a data warehouse. ELT, meanwhile, focuses on loading the data first and then transforming it within the data warehouse. The choice between ETL and ELT often hinges on the specific technologies in use, data volumes, and the desired performance outcomes (Moss & Atre, 2003).

Analytical Applications

These are specialized software solutions used to interpret and analyse business data. Analytical applications help in extracting meaningful patterns and insights from raw data, thereby aiding decision-making. They typically encompass capabilities ranging from basic statistical analysis to advanced predictive modelling. Their purpose is to turn raw data into actionable intelligence (Sharda, Delen & Turban, 2013).

Reporting

Reporting tools are essential to the BI infrastructure. They convert data into organized visual formats like tables, charts, and graphs, thereby simplifying the interpretation and decision-making process for stakeholders. These tools offer features like real-time reporting and customizable dashboards, enabling users to view data from different angles and levels of detail, depending on their specific needs (Few, 2006).

Data Mining

Data mining is the methodical exploration of large data sets to discern significant trends, relationships, and deviations. This discipline integrates elements of statistical examination, machine learning principles, and traditional database management. By employing these techniques, data mining attempts to reveal underlying trends and nuances often concealed within the sheer volume of data (Han, Pei & Kamber, 2012).

Information Management

In the realm of BI, information management refers to the practices and procedures that govern the handling of data. This includes ensuring the accuracy and quality of data, managing metadata (data about data), and enforcing security protocols to protect sensitive information. Effective information management is crucial because the reliability and security of BI insights are only as good as the data they are derived from (Loshin, 2010).

2.3 Self-Service Business Intelligence

Self-Service Business Intelligence (SSBI) represents a significant trend within the realm of Business Intelligence. These applications adhere to core BI principles while aiming to address a critical business challenge. The pivotal aspect is the need for flexibility and autonomy in data analysis by end-users of BI outputs (Pour et al., 2018).

In the book by Pour, Mariška, Stanovská and Šedivá (2018), titled “Self-service business intelligence: jak si vytvořit vlastní analytické, plánovací a reportingové aplikace”, the authors expand on the concept of SSBI, explaining how it enhances the traditional BI framework by enabling users to conduct personalized analysis and generate reports on the data accessed without the need for IT department intervention. This capability significantly broadens the autonomy, flexibility, and options for non-IT personnel, allowing them to quickly find answers to specific inquiries. Various studies support this notion by emphasizing that the timely availability of necessary information greatly impacts the accuracy of managerial decision-making. Although within an SSBI environment, the IT team continues to play a crucial role as the primary custodian of the organization's data, this role becomes even more critical here. It is essential to provide users with access to well-processed data in a comprehensible manner to ensure they can generate accurate reports and other BI outputs without requiring constant IT support.

According to Claudia Imhoff and Colin White (2011), the alteration in the role of IT within the SSBI environment engenders the following outcomes:

For the users: They are provided with data and the requisite access, enabling them to analyse and evaluate it anytime based on their current needs using the provided tools.

For the IT units: A significant amount of data work is moved to the users, saving IT department from being overwhelmed with constantly changing and new demands for reports and analysis. This simultaneously frees up the IT staff to concentrate on management and initiatives that increase the company's value.

For the organization at large: The shift culminates in heightened user satisfaction with the IT department, which is now perceived as a more adaptable ally capable of better catering to business requirements.

Self-Service Business Intelligence (SSBI) is widely discussed in various academic and business publications. One of the fundamental books on the topic is “Data Science for Business” by Foster Provost and Tom Fawcett (2013). Based on their insights and common practices in the field, below are the main recommendations when implementing an SSBI solution.

- **Understand the Business Context:** Before delving into the technicalities, it is crucial to understand the business problem, goals, and stakeholders.
- **Ensure Data Quality:** Before conducting any analysis, the data must be cleaned and validated to ensure its quality and reliability.
- **User-Friendly Tools:** It is important to select intuitive and user-friendly tools to allow non-technical users to explore and analyse data without extensive training.
- **Training and Support:** Even with user-friendly tools, users need some level of training. Ongoing support and resources should be provided to help them use the tools effectively.
- **Data Governance:** Strong data governance practices should be implemented. This includes defining accesses, maintaining data quality, and ensuring data privacy.
- **Iterative Approach:** SSBI is not a one-time process. It should be iterative, continually evolve and adapt, taking into account user feedback and the dynamic requirements of the business.
- **Collaboration**: Encourage collaboration among users. Sharing insights, visualizations, and reports can lead to a more comprehensive understanding of the data.

- **Security:** Ensure that data is secure. Implement role-based access, data encryption, and other security measures to protect sensitive information.
- **Scalability:** As the business grows, so will the data and user needs. Choose solutions that can scale with your organization.
- **Feedback Loop:** Provide feedback channels where users can report issues, request new features, and make suggestions for enhancements.

2.3.1 Procedure for Implementing SSBI Applications

For the successful implementation of SSBI applications within standard IT projects, it is advisable to adhere to the following phases: introductory study for SSBI, analysis and design of the SSBI application, implementation, and deployment of SSBI applications. However, it is important to note that the division of these phases can vary based on business practices or the chosen methodology (Pour et al., 2018).

Introductory study for SSBI

The phase emphasizes the importance for businesses to critically assess their existing analytical, planning, and reporting functions before initiating SSBI applications. SSBI solutions are distinctly categorized into those with a data warehouse and those without. For small companies, the need for a data warehouse may not be immediately evident, but for larger entities, it is essential. Integral steps in this preparatory phase include defining the SSBI objectives and expected outcomes, curating a user catalog, outlining specific SSBI requirements (both functional and relating to data sources), selecting an appropriate SSBI tool and its architectural framework, and culminating with the formulation of the project's schedule and economic attributes. The objectives and expected outcomes can vary according to different areas of management and can impact not only the financial aspects but also employee training, company-partner relations, and more. It is also important to understand the users of the SSBI solution, whether they are managers or regular employees. The number and structure of users can affect the outcome. Notable specifications could include seamless access to data sources, swift development and deployment, quality assurance, and a user-friendly interface (Pour et al., 2018).

Analysis and design of the SSBI application

The analysis and design of the SSBI Applications are pivotal for ensuring effective implementation of SSBI solutions. Given that SSBI solutions vary in scope and complexity, quality analysis and design are crucial for successful applications. Key objectives of this phase include a detailed specification of user requirements for the SSBI applications, ensuring alignment with the enterprise's needs while respecting existing BI functionalities. This phase also encompasses the analysis and design of the SSBI solution's content, including dimensions, indicators, their fundamental attributes, and interrelationships. Drawing insights from dimensional modelling, data models for SSBI solutions are crafted. Lastly, this phase culminates in the design of the necessary functionality for SSBI applications, detailing content specifics for individual analytical tables, dashboards, and reports (Pour et al., 2018).

Implementation of SSBI application

The primary goal of this phase is the creation, testing, and documentation of the desired SSBI applications, in line with the results of the analysis and design stages. The implementation process may vary significantly based on the chosen software tools, such as Spotfire, QlikView, Qlik Sense, Power Pivot and Power BI. A systematic approach is employed in this phase, which includes the creation of data model tables, defining relationships between individual tables, visual representation, and verification of the established links in the data model, performing necessary calculations, tests, and other operations using specific programming languages like DAX. Additionally, the phase focuses on generating contingency tables, contingency graphs over the data model, and implementing dashboards and scorecards of different types. One of the recommendations for this phase emphasizes the need for choosing appropriate software tools, ensuring their cost-effectiveness, and providing adequate training for users. Furthermore, the success of SSBI application implementation largely hinges on ensuring the availability and quality of data inputs into the SSBI databases, as this is often a significant limitation in SSBI application realization (Pour et al., 2018).

Deployment of SSBI application

The Deployment of SSBI Applications focuses on the real-world implementation of SSBI solutions. It involves preparing the necessary operational environment, which includes selecting appropriate software versions and deploying applications on both servers and user devices. This stage ensures that continuous data updates, termed 'refreshes', function seamlessly. Training is essential for users not initially involved in the application's creation. It's noteworthy that while SSBI grants users more autonomy in managing applications, it also demands refined expertise in analytical methods like dimensional modelling. Consequently, deployment is a crucial step in establishing the SSBI applications in a practical setting (Pour et al., 2018).

2.4 Power BI

Microsoft Power BI is a set of software services, applications, and connectors designed to aggregate disparate data sets into clear, visually engaging, and interactive analyses. It allows users to seamlessly integrate data, whether it originates from a mix of cloud and on-site data storage systems or just an Excel file. Power BI provides a platform for users to easily access their data, highlight significant points, and share this information to with a desired audience (Microsoft, 2023).

Power BI can be used for a variety of business analytics needs, allowing users to pull in data from different sources and convert it into a meaningful analysis through easy-to-understand visuals. Power BI is composed of various interconnected components, the fundamental three being datasets, reports, and dashboards. (Ferrari & Russo, 2016).

At its core, Power BI is designed to deliver invaluable business value by offering a range of features that facilitate data cleaning, transformation, anomaly detection, trend spotting, and dashboard creation, among other capabilities. It empowers users to consume reports and dashboards, thereby fostering a data-driven culture within organizations. The tool's ability to utilize AI insights for detecting anomalies and identifying trends is a proof of its advanced analytics capabilities, providing a platform for enhanced data visualization and informed decision-making processes (Microsoft, n.d.).

Its cloud-based nature facilitates collaboration, with the Power BI service being the online SaaS (Software as a Service) aspect of Power BI that works hand in hand with Power BI Desktop for creating reports and with Power BI Mobile for consuming them (Powell, 2018). In an increasingly data-driven world, tools like Power BI are paramount for businesses to make informed decisions. Power BI is structured around various components. Each component serves a specific role within the broader Power BI ecosystem.

- **Power BI Desktop**: This is a Windows application used primarily for designing and publishing reports. It provides a rich interface for data modelling, creating measures, and designing interactive reports. Data analysts and other professionals typically use this application to transform raw data into meaningful insights (Ferrari & Russo, 2016).
- **Power BI Service:** Often referred to as “Power BI online”, this is a cloud-based service where the reports and dashboards created in Power BI Desktop are usually published for wider consumption. It also provides capabilities like Q&A, which allows natural language querying of the data (Powell, 2018).
- **Power Query:** An ETL (Extract, Transform, Load) tool, Power Query enables users to connect, transform, and enhance their data. It supports various data sources, from traditional databases to web services, and provides a user-friendly interface for data transformation tasks (Powell, 2018).
- **Power BI Mobile:** This is a mobile application that lets users access reports and dashboards on the go. It is available on various platforms like iOS, Android, and Windows (Microsoft, 2023).
- **Power BI Gateway:** It is a bridge that facilitates fast and secure data exchange between the Power BI service and on-premises data. This ensures data refreshes are timely, and reports are up to date (Rad, 2018).
- **Dataflows:** These are ETL processes that can be defined within the Power BI Service. They allow users to connect to, transform, and load data into Azure-based data lake storage, from which datasets can pull the data (Microsoft, 2023).

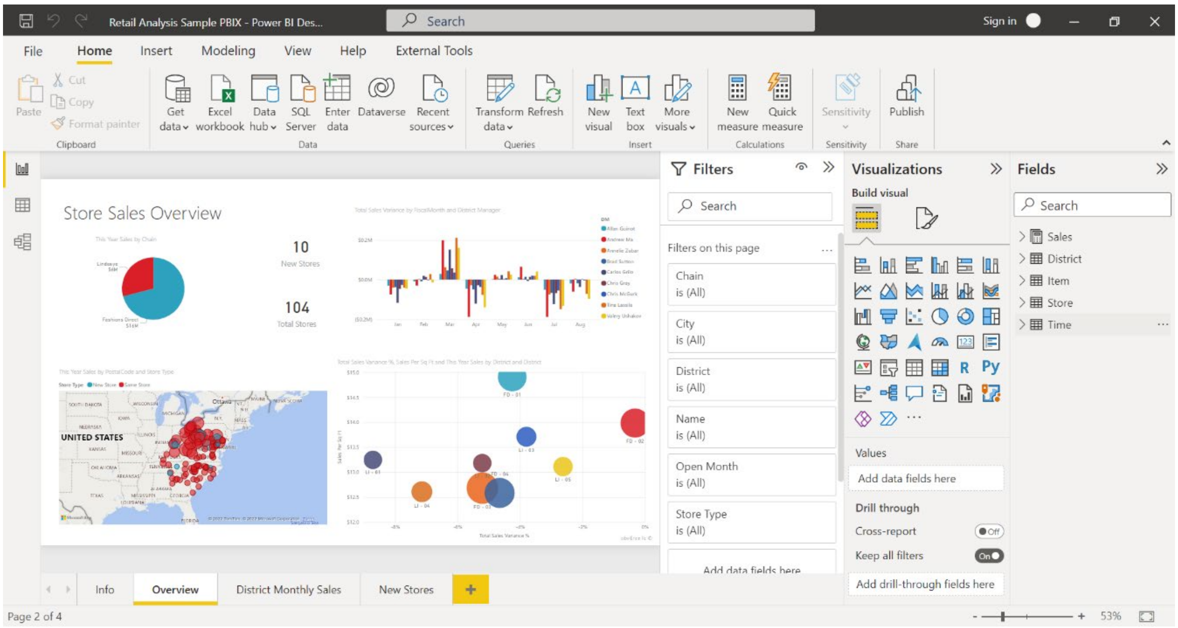


Figure 2: Power BI Desktop interface. Source: Microsoft.

2.4.1 Dashboards and Reports

In Power BI, dashboards and reports serve as mediums for visualizing and interacting with data. The distinction between them lies in the level of detail and interactivity they offer. A dashboard provides a high-level overview, aggregating key metrics and insights on a single canvas. It can pull in visualizations from multiple reports and datasets, enabling a bird's eye view of critical business indicators. Conversely, a report consists of one or more pages of visualizations, offering a more detailed analysis and allowing users to explore data interactively. Both dashboards and reports are instrumental in conveying data stories, albeit serving different purposes based on the depth of analysis required (Microsoft, 2023).

2.4.2 Visualization

Power BI supports different visualization types, including charts, graphs, maps, and matrices, amongst others. The choice of visualization greatly impacts how effectively data insights are conveyed to the audience. In the book "Storytelling with Data: A Data Visualization Guide for Business Professionals" Cole Knaflic (2015) discusses numerous lessons on enhancing data visuals. Below are a few general recommendations for designing dashboards and creating visualizations:

- **Audience Consideration:** Tailor the dashboard layout to the audience's needs and usage scenarios. Understand the key metrics that aid decision-making for the audience and ensure that these are prominently featured on the dashboard.
- **Clarity and Simplicity:** Strive for a clean, uncluttered dashboard that displays only essential information. Avoid visualizations that are hard to read, like 3D charts, and opt for bar or column charts for comparing values, as they are easier to interpret.

- **Visualization Selection:** Choose visualizations that paint a clear picture of the data. For instance, bar charts are suitable for comparisons, line charts for showcasing trends, and scatter plots for depicting relationships.
- **Consistency:** Maintain consistency in chart scales, colour schemes, and dimension values within charts to avoid confusion.
- **Performance Optimization:** Optimize the performance of the report to ensure a smooth user experience, especially when dealing with large datasets (Knaflic, 2015)

2.4.3 Dashboard Types

According to Wayne W. Eckerson's book titled "Performance Dashboards: Measuring, Monitoring, and Managing Your Business" (2012), dashboards are categorized into three types: operational, tactical, and strategic. Eckerson explains them as follows:

- **Operational Dashboards** concentrate on real-time or close to real-time monitoring of operational processes, events, and activities. They are crucial for front-line workers to manage and control operational processes with detailed data.
- **Tactical Dashboards** are designed for analysing and measuring departmental activities, processes, and goals. They enable executives to review and benchmark performance across the company and allow managers to monitor and optimize processes.
- **Strategic Dashboards** aim at tracking progress towards achieving strategic objectives, often employing methodologies like the Balanced Scorecard. They are primarily used by executives to communicate strategy and review performance, typically in monthly strategy or operational review meetings.

Understanding the dashboard's type helps in designing it with the right features and functionalities that suit its intended use. Operational dashboards require real-time data updates, while strategic ones might emphasize long-term trends and goals. The choice of metrics and data presentation also varies. Operational dashboards might track immediate performance indicators, whereas strategic dashboards might focus on high-level KPIs. Another benefit is the correct identification of primary users. Operational dashboards are often used by frontline staff, tactical by middle management, and strategic by senior executives. Moreover, each type of dashboard requires a different approach to data visualization and presentation, tailored to the needs and decision-making processes of its users (Eckerson, 2012).

The insights gained from this section were instrumental in identifying the type of the proposed dashboard that best aligns with the company's specific requirements (refer to Section 3.2).

3 Introductory Study

As highlighted in Section 2.3.1 and based on the insights from Pour et al. (2018), it is imperative to thoroughly evaluate the existing analytical, planning, and reporting capabilities of a company prior to implementing SSBI applications within IT projects. This chapter provides an introduction to the company, delineates the objectives and anticipated outcomes, offers an overview of the company's processes, and delves deeply into the company's current state of performance assessment.

3.1 Company Profile

To meet the primary objective of this thesis, which is the design and implementation of an SSBI solution that visualizes and analyses the performance of IT service suppliers, it is essential to review the current state of reporting and pinpoint the challenges within the process.

The company under analysis has chosen to remain anonymous and will be referred to as "XYZ" throughout this document. XYZ operates across various domains within the healthcare and life sciences sectors, marking its global footprint by developing, manufacturing, and marketing a diverse range of products. With operations in multiple countries and a substantial workforce, XYZ is categorically positioned as a large enterprise.

The company has its own IT department, however for various specialized IT services it relies on external suppliers from different countries. These services are governed by "Underpinning Contracts" (UC), as defined by the ITIL (Information Technology Infrastructure Library) framework. ITIL describes an Underpinning Contract as an agreement between the IT service provider and a third-party vendor. The third-party offers goods or services that support the delivery of an IT service to a customer (ITIL, 2019). In the context of XYZ, these are the contracts between the IT department and external IT service suppliers. Furthermore, these Underpinning Contracts are often governed by Service Level Agreements (SLA), ensuring that the third-party vendors meet the expected service delivery standards. SLAs are the agreements between the company's IT department (which could be considered the "service provider" even if some services are outsourced) and its internal employees (the end-users). The SLA states what the employees can expect in terms of IT service quality, reliability, response times, etc. To ensure cohesion and streamlined operations internally, Operational Level Agreements (OLA) are also pivotal. Those are the agreements detailing how different internal departments or teams cooperate to deliver the services promised in the SLA (ITIL, 2019).

In this study, the focus will be on the Underpinning Contracts (UC) established between the company's IT department and external IT service suppliers. The company engages with four distinct IT service suppliers, all of whom operate under the framework of UC agreements.

Consequently, the assessment and further sections will concentrate on the performance and deliverables of these four suppliers and the Underpinning Contracts.

3.2 Objectives and Expected Outcomes

The primary focus of this thesis is the creation of a Self-Service Business Intelligence (SSBI) solution to analyse IT service supplier performance within the company. This overarching goal is supported by two key sub-objectives: automation of service level reporting and enhanced visibility. The aim is to transition from reliance on externally provided data to utilizing internally generated, automated reports, enhancing the consistency of supplier performance insights. Regarding the second sub-objective, the intention is to increase transparency in supplier performance, tailored to meet the requirements of internal teams and suppliers, and refined based on feedback from solution testing.

Upon its implementation, the company is set to benefit from a dashboard for daily use by both employees and suppliers. This tool will automate service level reports and provide insights into performance trends. Additionally, this study presents an overview of the current reporting landscape, outlines user requirements, highlights testing outcomes, and pinpoints potential areas for improvement.

As outlined in Section 2.4.3 in the context of Eckerson's framework, dashboards are categorized into operational, tactical, and strategic types. Considering the objectives and expected outcomes described, the proposed SSBI solution falls under a hybrid category, embodying characteristics of both operational and tactical dashboards. It blends the daily monitoring and control characteristics of operational dashboards, as seen in the objective to focus on automating service level reporting. Simultaneously, it incorporates the analytical and managerial oversight typical of tactical dashboards, demonstrated by enhancing visibility and tailoring insights based on feedback. Therefore, this dual functionality aligns the proposed dashboard with both operational and tactical aspects, classifying it as a hybrid type.

3.3 Description of Monitored Service Levels

Service level, often articulated in a Service Level Agreement (SLA), specifies a particular standard that a service provider commits to meeting. It is a measurable performance metric related to the delivery of a specific service. Service levels ensure that there is a clear understanding and agreement between service providers and their customers about what to expect in terms of service delivery. Non-compliance with these standards may lead to penalties or other contractual implications (ITIL, 2019)

Service delivery within the company largely relies on IT support in the form of setting up and resolving tickets within the designated ticketing tool. Tickets can be categorized as incidents, service requests, changes, problems, and more. The definitions and governance of these categories stem from the IT Service Management (ITSM) guidelines, which are supported by the ITIL framework.

ITSM encompasses various processes such as Incident Management, Problem Management, Change, Release, and many others. For the purpose of this thesis, the primary focus will be directed towards “incident tickets” and the associated “Incident Management” process. This will be elaborated upon in the subsequent section.

A review of the company's current reporting framework reveals that distinct service levels have been established for suppliers. These service levels are mainly determined by the types of services suppliers provide, such as application, network, security, service desk, on-site Support, and other infrastructure services. It is important to note that these service levels are further differentiated by ITSM processes they correspond to.

The importance of certain service levels varies based on the service's role, its impact on end users and its relevance to the company's strategic goals. These are called “Critical Service Levels” (CSLs) and their measurable counterparts are “Key Performance Indicators” (KPIs). Critical Service Levels are a set of performance indicators that are vital for the operation of the business. Any deviation or failure to meet these service levels could lead to a significant negative impact on the company’s operations, such as loss of revenue, damage to reputation, or legal issues (ITIL, 2019). In XYZ, failure to meet these service levels leads to financial penalties or additional contractual consequences for suppliers.

KPIs, on the other hand, are metrics used to evaluate the effectiveness and efficiency of the IT service process. They are essential for long-term service quality and continuous improvement. KPIs encompass a broader range of metrics, such as overall customer satisfaction, the percentage of incidents resolved within the first contact, and the average time taken for repairs (Hiles et al., 2016).

The approach for measuring Service Levels discussed in this chapter is derived from data outlined in internal company documentation. Table 1 offers a representative illustration of these Service Levels. It is important to note that this table is not exhaustive and only offers a selection of the various Service Levels monitored. Additionally, the target figures have been adjusted and do not reflect the actual targets set by the company, ensuring the confidentiality of internal benchmarks.

Table 1: Sample overview of monitored Service Levels for IT Services. Source: Author's own creation based on internal company materials.

Service	Process	Service Level	Type	Measurement Period	Expected Target	Minimum Target
Application Services	Incident Management	Resolution Time for Critical Priority Incidents	CSL	Monthly	96.00%	94.00%
		Resolution Time for High Priority Incidents	CSL	Monthly	96.00%	94.00%
		Resolution Time for Moderate Priority Incidents	KPI	Monthly	96.00%	94.00%
		Resolution Time for Low Priority Incidents	KPI	Monthly	96.00%	94.00%
		Response Time for Critical Priority Incidents	CSL	Monthly	99.00%	97.00%
		Response Time for High Priority Incidents	CSL	Monthly	99.00%	97.00%
		Response Time for Moderate Priority Incidents	KPI	Monthly	99.00%	97.00%
		Response Time for Low Priority Incidents	KPI	Monthly	99.00%	97.00%
	Service Request Management	Resolution Time for Service Requests	CSL	Monthly	93.00%	90.00%
		Response Time for Service Requests	CSL	Monthly	93.00%	90.00%
	Change Management	Change Requests Completed On-Time	CSL	Monthly	95.00%	93.00%
		Change Requests Closed On-Time	CSL	Monthly	95.00%	93.00%
Infrasturcture Services	Incident Management	Resolution Time for Critical Priority Incidents	CSL	Monthly	96.00%	94.00%
		Resolution Time for High Priority Incidents	CSL	Monthly	96.00%	94.00%
		Resolution Time for Moderate Priority Incidents	KPI	Monthly	96.00%	94.00%
		Resolution Time for Low Priority Incidents	KPI	Monthly	96.00%	94.00%
Infrasturcture: Database Services	Incident Management	Response Time for Critical Priority Incidents	CSL	Monthly	98.00%	96.00%
		Response Time for High Priority Incidents	CSL	Monthly	98.00%	96.00%
		Response Time for Moderate Priority Incidents	KPI	Monthly	97.00%	94.00%
		Response Time for Low Priority Incidents	KPI	Monthly	97.00%	94.00%

Note: Targets have been adjusted to preserve confidentiality.

3.4 Incident Management

Incident management in the context of IT Service Management (ITSM) is a process which aims to quickly restore normal service operations while simultaneously lowering the impact on business functions. This approach is essential for maintaining high levels of service quality and availability. An "incident" is typically defined as an unplanned interruption to an IT service or a reduction in the quality of an IT service (ITIL, 2019).

Within the company, incidents are classified into different priorities based on their impact and urgency. Priorities are categorized as Critical, High, Moderate, and Low. Each category has a different target resolution or response time, with more critical incidents requiring a faster response and resolution.

Resolution Time for Incidents

The service level referred to as “Resolution Time for Incidents” in IT Service Management (ITSM) is a measure of the time required to address and resolve an incident from the moment it is reported. This metric is a critical part of the Underpinning Contract (UC) between a service provider and its customers or users. The resolution time starts when an incident is reported and ends when the incident is closed to the satisfaction of the user. It is important to mention that “resolution” means that the user is able to return to normal operation, even if it's through a workaround rather than a permanent fix.

This metric can be compared against the targets established in the Underpinning Contracts to determine if service levels are being met. The measurement of the resolution time for incidents involves tracking the percentage of incidents resolved within the UC to assess their performance.

$$\%\ of\ Incidents\ Resolved\ within\ UC = \left(\frac{Number\ of\ Incidents\ Resolved\ within\ UC\ Target}{Total\ Number\ of\ Incidents}\right) \times 100$$

To calculate this for incidents of different priorities and services, it is necessary to segment the incidents based on their priority level and service and then apply the formula to each segment:

$$\%\ of\ Incidents\ Resolved\ within\ UC = \left(\frac{Incidents\ within\ Target_{of\ a\ Priority\ and\ Service}}{Total\ Incidents_{of\ a\ Priority\ and\ Service}}\right) \times 100$$

Response Time for Incidents

The “Response Time for Incidents” service level in IT Service Management (ITSM) is another critical measure that indicates the time taken for the service provider to initially respond to an incident. Contrary to resolution time, response time does not consider the total time taken to resolve the incident. Instead, it is focused on the time taken to acknowledge an incident and provide the first response to the user. This metric reflects how quickly a service team engages with an incident after it has been reported.

The response time is defined within the Underpinning Contract (UC) and can vary depending on the priority level of the incident. For example, the expected response time for a critical incident is usually shorter than for a low priority incident.

The formula for tracking the percentage of incidents that met the first response time as stipulated in the UC is as follows:

$$\%\ of\ Incidents\ Responded\ within\ UC = \left(\frac{Number\ of\ Incidents\ Responded\ within\ UC\ Target}{Total\ Number\ of\ Incidents}\right) \times 100$$

This calculation helps support teams understand their effectiveness in acknowledging and beginning the incident management process. Maintaining quick response times is often a key component of customer satisfaction and an important performance metric for service teams.

3.5 Incident Lifecycle

The Incident Ticket Lifecycle in the context of IT Service Management (ITSM) under the framework of ITIL is a structured process that describes the various stages an incident ticket goes through from the time it is created until it is resolved and closed. Underpinning Contracts (UCs) and Service Level Agreements (SLAs) play a significant role throughout this lifecycle.

The lifecycle begins when an incident is identified, either by a user who reports it or through an automated alert from a monitoring system. Every incident is logged as a ticket in the company's ticketing tool. The ticket contains all the details of the incident, including a unique identifier, description, time of report, the reported impact, and a preliminary categorization. The incident is categorized and prioritized based on its nature, urgency, and impact on the business. Prioritization helps in determining the response and resolution times as defined by the UC. The service desk undertakes an initial diagnosis to attempt to resolve the incident using available knowledge articles or known errors.

Underpinning Contracts establish targets for various aspects of incident management, including both the response and resolution times of the incidents that are specifically assigned to the suppliers. These targets are typically tiered according to the priority of incidents. For example, Critical priority incidents may have a response time of within 1 hour and a resolution time of within 4 hours. On the other hand, High priority incidents may have a response time of within 2 hours and a resolution time of within 8 hours. If an incident is not managed within the agreed target times, it is considered a breach of the UC. Breaches can result in penalties for the supplier, depending on the terms of the UC, and often lead to customer dissatisfaction.

When the incident cannot be resolved at the initial stage, it is reassigned to more specialized support teams based on categorization and prioritization. In such cases, each internal or supplier team might be responsible for a specific component or aspect of the IT services. Effective resolution of such incidents typically requires collaboration between different support teams. When different suppliers are involved, UCs with each supplier will dictate their specific responsibilities and performance targets, including how they engage in cross-supplier incidents. Once a solution or a workaround is found, actions are taken to restore service to the user. After the resolution of the incident and subsequent confirmation from the user, the ticket is then closed. A closure code and any relevant notes are added to the ticket.

An incident is classified as a Major Incident when it affects a significant number of users, poses a serious impact on business operations, and requires an urgent response. Due to their severity and potential repercussions, major incidents are subject to distinct resolution targets and management protocols.

An incident task can be considered a subtask that is part of the process to resolve an incident. During the course of managing an incident, various tasks may be created to track the work that needs to be completed. An example of this can be when an incident may require an initial investigation, a hardware replacement, and a software reconfiguration to be resolved. Each of these steps could be set up as individual incident tasks within the company's ticketing tool. These tasks are usually assigned to different individuals or groups who have the specific skills required to complete them. The incident remains open until all associated tasks are completed and the service is restored.

Incidents and their adherence to UC targets are important for the continuous improvement of IT services. Analysing incidents and the success rate of meeting UC targets can provide insights into service weaknesses, leading to improvements in IT processes, infrastructure, and support capabilities. UCs can include penalties for failing to meet the agreed targets. These penalties serve to incentivize suppliers to prioritize incidents that they are responsible for, in line with the agreed terms.

3.6 Current State of Performance Assessment

At present, the assessment of IT service supplier performance primarily relies on reports generated from the company's ticketing tool. These reports are saved in Excel formats, allowing for further analysis conducted by the suppliers. This method involves filtering incident tickets using various criteria, identifying incidents relevant to the supplier's service offerings, and categorizing them according to their priority levels.

When discussed with the supplier management team and internal stakeholders, several challenges have been identified with the current assessment methodology:

- **Ticketing Tool Limitations:** The ticketing tool only displays the final resolving support team for an incident. Consequently, to understand all the involved groups who contributed to the resolution of the incident, one must open each ticket individually. This makes it difficult to identify and monitor cross-supplier incidents that involve multiple suppliers or internal groups.
- **Lack of Standardization:** With multiple suppliers, each supporting various IT services, there is no uniform approach for report extraction and monitoring, resulting in the absence of a standardized method of performance assessment.
- **Extraction Constraints:** The ticketing tool has a set extraction limit of 10,000 rows, which, considering the size of the organization and the number of tool users, is very restrictive.
- **Service Level Limitations:** With the current underpinning contracts (UCs) created in the ticketing tool, it is impossible to display incidents in alignment with the agreed-upon service levels. As a result, specific calculations for these service levels are performed in Excel.
- **Visualization Challenges:** The ticketing tool's built-in capabilities for calculations and graphical representation fall short of providing a comprehensive view of overall performance.
- **Reliance on Static Tools:** Analyses carried out in Excel or CSV formats post-extraction are static, lacking the ability to be updated automatically.

- **Presentation Hurdles:** The absence of a consolidated report complicates the process of navigating various reports during performance review meetings, slowing down decision-making.
- **Supplier Division Absence:** The ticketing tool does not incorporate a distinct field for supplier division, requiring manual selection within the tool's filters for accurate identification.
- **Data Trustworthiness Concerns:** The supplier management team, responsible for assessing supplier performance, often depends on data provided by the suppliers themselves. Given that suppliers extract this data from the company's ticketing tool, there is a potential risk regarding the reliability of such numbers. There is an inherent concern that suppliers may underreport or omit incidents that have breached agreements, particularly when they face potential penalties for such breaches.
- **Time-Consuming:** Significant time is required from suppliers and internal stakeholders to manually sift through incident and incident task reports due to the above-mentioned limitations.

3.7 User Catalog

The user catalog for the supplier performance dashboard, integral to the SSBI solution, encompasses a triad of key user groups: internal IT employees, external IT service providers (suppliers), and the supplier management team.

Internal IT Employees: This group comprises personnel within the organization who are actively involved in the operational aspects of IT service delivery. They utilize the dashboard to monitor ongoing incidents, track service level compliance, and ensure that the IT services are aligned with business needs. The insights gleaned from the dashboard enable employees to make informed decisions and take timely actions to maintain service excellence.

Suppliers: These are the external third-party IT service providers, whose service delivery and performance are under scrutiny within the SSBI solution. They interact with the dashboard daily to review their service levels, analyse performance metrics, and identify any service breaches that require attention. The dashboard serves as a transparent medium for suppliers to self-assess their performance and drive continuous improvement.

Supplier Management Team: This team is tasked with the oversight and governance of supplier performance against the given underpinning contracts. They rely on the dashboard for an aggregated view of supplier performance, enabling them to evaluate compliance with contractual obligations, examine service level breaches, and conduct performance reviews. The dashboard facilitates a factual and objective basis for dialogue with suppliers, fostering a collaborative approach to performance management and enhancement.

Each user group engages with the SSBI solution tailored to their specific needs and roles, creating a comprehensive environment for the management and optimization of supplier performance based on concrete data and analytics derived from the company's ticketing tool.

3.8 Data Source Retrieval

This section focuses on the method employed for sourcing and retrieving data for the SSBI dashboard. Incident reports are generated in the company's ticketing tool, and underpinning contracts (UC) are embedded within to ensure alignment with service delivery expectations. Contracts are established in advance encompassing pre-agreed terms that stipulate target durations for incident resolution and response times to which suppliers are expected to adhere. However, there is no direct interface to connect a Self-Service Business Intelligence (SSBI) tool with the ticketing tool to access this data.

Given the lack of a direct data connection, the company leverages a cloud-based solution as a system of reference. The solution consists of different data sources one of which being the ticketing tool, then a data lake and a data warehouse. Each department in the company has its own data mart which is a tailored subset of the data warehouse. Within this framework, a direct connection is configured between the SSBI tool and the data warehouse, where the data marts are situated. By connecting to a specific subset of the data warehouse, the dashboard has direct access to the most relevant and updated datasets, which are refreshed on a daily basis. This ensures that the dashboard reflects the latest data, giving stakeholders timely insights into supplier performance.

In order to facilitate the connection between the database and any BI tool, a non-personal account (NPA) is required. These accounts, operating within an SQL database framework set up in a secure cloud environment, are endowed with specific permissions necessary for accessing the data mart. For the purpose of this thesis, an NPA account was created, and the requisite privileges to interact with the production environment of the database were duly authorized. This setup allows for a seamless and secure connection, enabling the SSBI tool to retrieve the necessary data for comprehensive reporting and analysis.

The data retrieval process for the dashboard, following a comprehensive analysis, involves extracting information from several key tables within the database and the company's SharePoint site. These tables, selected for their relevance and utility for the dashboard, include:

- **Incident Table:** This central table captures comprehensive details about each incident. It includes information on the incident's status, opening and closing dates, the presence of major incidents, and various other specifics recorded by the ticketing tool. Additionally, it tracks the number of times an incident has been reassigned or reopened and whether it qualifies as a major incident.
- **UC-Incident Resolution Table:** This table is a crucial addition to the database, combining details related to the Underpinning Contracts (UCs) for incident resolution. It provides insights into whether the resolution targets for tickets were met or exceeded. This table is particularly significant for historical analysis, as it shows all the changes that occurred in the incident including any changes in ticket priority.
- **Ticket SLA Table:** This table encompasses the SLAs, OLAs, and UCs associated with various ticket types, including incidents, requests, problems, and changes. In the context of this thesis, it is utilized to extract data specifically related to the

incident response UCs. It focuses on current SLAs and includes details about the last resolver group and other relevant SLA-related information, without historical incident data.

- **Support Groups Table:** This table documents detailed information about each support group within the organization, covering both supplier and internal groups. It includes specifics such as group names, hierarchical positioning, group types, manager details, and a general description of each group. Additionally, it provides information regarding the current status of each group, indicating whether the group is active or not.
- **Group Mapping:** Another data source originates from the company's SharePoint site, where a manually updated Excel file is maintained. This file is crucial as it contains a few specific mappings of supplier support groups to the corresponding company divisions and services that should be updated in the ticketing tool. This file ensures accurate alignment in the data analysis.

The dataset has been modified to ensure the confidentiality of sensitive information. In this regard, the names of suppliers and their respective support groups have been anonymized. For instance, original support group names have been replaced with generic identifiers such as 'Support Group 1', 'Support Group 2', and so forth. Similarly, ticket numbers have been altered by appending random numerical characters, further ensuring data privacy. Moreover, based on user requirements, the scope of the data sources has been specifically confined to the year 2023 providing a focused and relevant dataset for analysis.

3.9 SSBI Tool Selection

When selecting an SSBI tool for implementing the supplier performance dashboard as part of this thesis, Microsoft Power BI was chosen for several reasons, both practical and strategic. According to Gartner's Magic Quadrant for Analytics and Business Intelligence Platforms, Power BI is often recognized as a market leader due to its comprehensive capabilities, ease of use, and integration with Microsoft products (Schlegel et al., 2023).

One of the key reasons for choosing Power BI was the supplier management team's familiarity with the tool. Their experience with Power BI allows continuous support of the dashboard. Consequently, this reduces the learning curve and accelerates the adoption of the new dashboard. Familiarity with the tool's interface and functionality can streamline the transition and promote more efficient use of the dashboard.

Other departments within the company were already using Power BI to analyse other ITSM processes and SLAs. This cross-departmental adoption indicates that Power BI has been vetted and accepted across the organization, which can facilitate inter-departmental data sharing and reporting consistency.

There was an existing tested connection from the company's database to Power BI. This infrastructure foundation allows for seamless integration and reliable data flow, which is crucial for up-to-date reporting and analytics.

4 Analysis of User Requirements

To accurately capture the essential requirements for a dashboard designed to monitor supplier performance in compliance with underpinning contracts, a series of meetings were conducted. These sessions took place both in person and virtually via the MS Teams application, engaging internal employees, the supplier management team, and the suppliers themselves. These discussions, informed by the insights into the current state of reporting as detailed in Section 3.6, led to the identification of user requirements. These requirements are systematically catalogued in Tables 2 and 3.

Table 2 encapsulates the business and data requirements—these articulate the specific types of information users need to access through the dashboard to facilitate decision-making and performance evaluation. On the contrary, Table 3 focuses on the operational aspects, delineating the functional requirements that define the dashboard's necessary behaviours and interactions.

The collection of user requirements was achieved through User Story Mapping, a technique that gathers concise and straightforward narratives of desired features from the user's perspective. According to Cohn (2004), user stories are typically phrased as follows: “As a [type of user], I want [an action] so that [a benefit/a value].” This format not only underscores the feature's utility but also its value proposition to the user (Cohn, M., 2004).

Each user story within the tables is prioritized as high, medium, or low, indicating the relative importance and urgency of the corresponding feature to the user's needs. This prioritization aids in the strategic development and phased rollout of the dashboard functionalities.

Table 2: Business and data requirements for Supplier Performance dashboard. Source: Author

User Story	**Priority**
As a service lead, I want to have a reporting tool, where I can assess supplier performance independently and not rely on suppliers for this information.	High
As a stakeholder, I want an interactive dashboard where I can filter incident tickets based on a combination of suppliers, priorities, support groups, and services for targeted analysis.	High
As a service lead, I want to view the total number of incidents closed within a specific month to assess overall service resolution efficiency.	High
As a performance manager, I want to see how many incidents each supplier, or specific supplier group has resolved and how many they have failed to evaluate their performance against contractual obligations.	High

As a supplier, I want to see incident resolution and response service levels specifically associated with the services I provide, and compare the percentage of successful tickets to agreed targets so that I can ensure compliance and identify service gaps.	High
As a service desk analyst, I want to see the number of tickets resolved for each priority level to ensure we are meeting our service level agreements.	High
As a performance analyst, I want to distinguish and compare the tickets resolved by a single supplier from those requiring cross-supplier collaboration to identify areas for process improvement	High
As a performance analyst, I want to view a list of all support groups involved in the resolution of each ticket, to analyse performance and efficiency across different teams.	High
As a service lead, I want service levels for incident response time to be calculated based on the first resolver group of each service for accurate measurements.	High
As an IT operations lead, I want to track major incidents separately to prioritize response strategies and resource allocation.	Medium
As a service lead, I want to track tickets that have undergone a change in priority (from low to medium for instance), to analyze shifts in urgency and resource deployment.	Medium
As a performance analyst, I want to see which tickets were resolved by individual support groups versus those requiring multiple groups, whether within one or several suppliers, to understand collaborative dynamics.	Low
As a performance analyst, I want to see service request, problem and change metrics displayed in the dashboard for a complete overview.	Low
As a supplier relationship manager, I need to identify tickets resolved by suppliers that resulted in user dissatisfaction, to enhance service quality and address specific issues with our suppliers.	Low

Table 3: Functional requirements for operational aspects of the Supplier Performance dashboard. Source: Author

User Story	Priority
As a supplier, I need the dashboard to refresh automatically on a daily basis so that I am consistently up to date with my performance metrics and can proactively manage service delivery.	High
As a supplier performance lead, I need the reports that display supplier deliverables (service levels) to be fully automated, ensuring data reliability and up-to-date performance tracking for effective service management.	High
As a supplier performance lead, I need a separate tab or report within the dashboard that enforces security constraints, ensuring some suppliers can only view their own service levels. Additionally, it is crucial that internal IT employees have the ability to see all suppliers' data for comprehensive oversight.	High

5 Design and Development of Dashboard

This chapter outlines the creation of a new dashboard for assessing supplier performance, developed through an agile-centric approach. This methodology emphasized iterative development cycles, allowing for continuous refinement based on stakeholder feedback. The process was guided by open communication with company stakeholders, ensuring the dashboard's alignment with their needs and the strategic objectives of the Self-Service Business Intelligence (SSBI) solution. This approach fostered a dynamic and responsive development environment, which was crucial for the practical and effective implementation of the dashboard.

5.1 Data Modeling

Figure 3 showcases the data architecture of the SSBI solution. The data model, conceptualized within Power BI, illustrates the integral tables and their interrelations, forming the structural foundation for data analysis and visualization within the dashboard. This model was created based on the user requirements mentioned in Chapter 4.

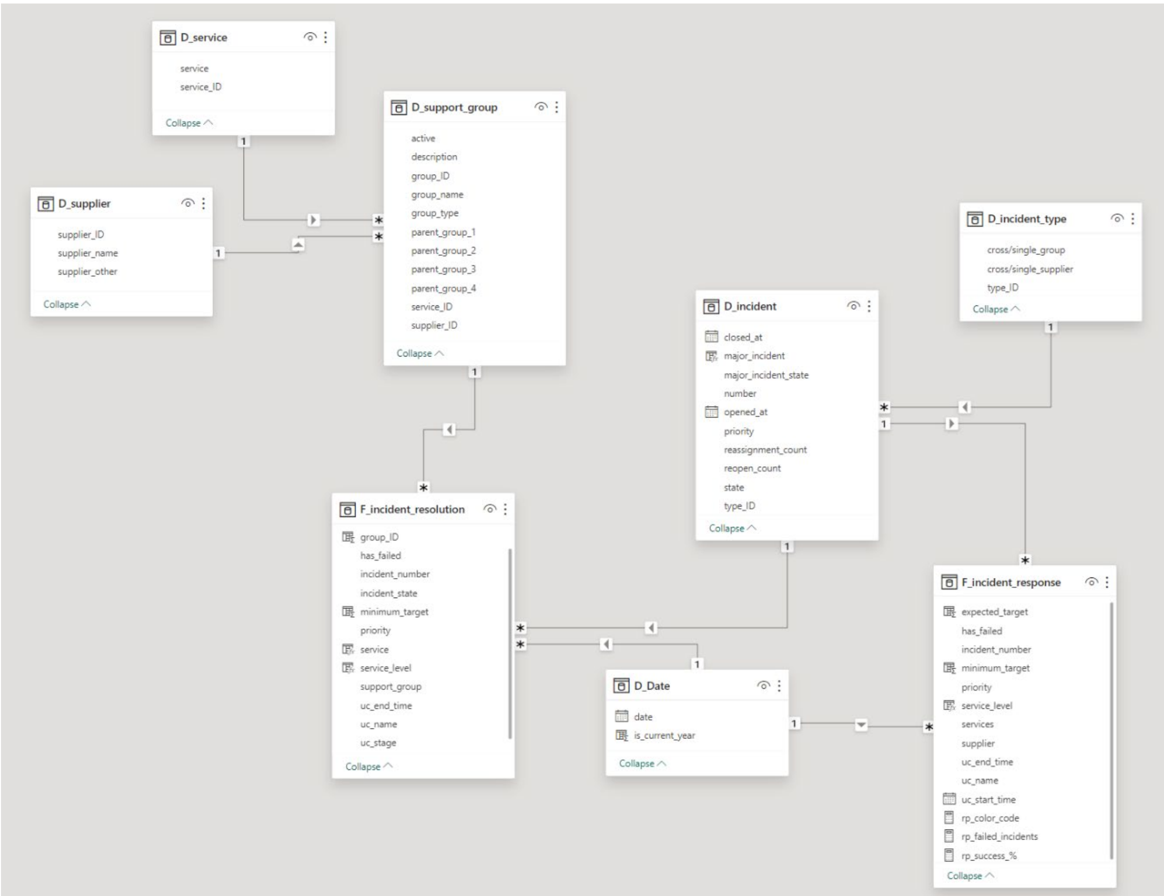


Figure 3: Data Model in MS Power BI. Source: Author.

After establishing the required connections with the company's database and the SharePoint site in Power BI Desktop, as detailed in Section 3.8, the data tables underwent transformations in Power Query before being loaded into the model. During the initial connection phase, specific tables were pre-filtered using SQL scripts to eliminate extraneous data rows, enhancing efficiency and focus. Some of these scripts are detailed in the Appendices for reference.

Subsequent to the connection setup, the data was further processed in Power Query. Common steps applied across all tables included automatic data type detection, which involved identifying the appropriate data types for each column. Additionally, this stage involved navigating to the appropriate tables within the data source relevant for analysis. The key to the data transformation process was the selective removal of columns. Simultaneously, certain columns were renamed to more practical and descriptive titles, facilitating easier understanding and manipulation in the ensuing stages of dashboard development.

In the end, after completing the data analysis and verifying data accuracy in the visualizations, a custom column was added to several tables to assign a random number to each row. Following the assignment of these random numbers, the tables were filtered to retain only a select number of rows, determined randomly. This step was undertaken as a measure to enhance confidentiality. It ensured that the images presented in this thesis maintained the integrity of the analysis while adhering to confidentiality requirements.

5.1.1 Dimension Table Incident

The “D_incident” table in the data model encompasses comprehensive details for each recorded incident in the ticketing tool. Central to this table is the unique “number” identifier for each incident, acting as the primary key that facilitates linkages to fact tables. Essential attributes contained within this table include the incident's current state, the presence of major incidents, the priority it was closed with, and the timestamps of when incidents were opened and closed. Moreover, it shows the number of times an incident has been reassigned and reopened.

The original dataset for the “D_incident” table is sourced from a database table, which catalogues all incident numbers along with their associated tasks and groups. Each incident is methodically mapped to various groups that were engaged in its resolution, as well as to the resolution of related tasks. To align with analytical objectives and user specifications, an SQL script was employed to select only incidents with a “Closed” state that opened from January 1st, 2023, onward.

Following this selection, the incidents were grouped in Power Query using the “All Rows” operation and columns were added to calculate the count of unique groups as well as unique suppliers linked to each incident number (example illustrated in Code 1). The data was then evaluated to determine whether an incident's resolution was the effort of a single group or multiple groups, and whether it involved a single supplier or multiple suppliers (Code 2). Each incident is then assigned a corresponding type_ID based on these determinations and unnecessary columns are removed.

Code 1: M code to count distinct groups associated with an incident. Source: Author.

```
= Table.AddColumn(#"Grouped Rows", "DistinctGroupsCount", each let
    IncidentData = [GroupedData],
    DistinctGroups = Table.Distinct(IncidentData, {"support_group"}),
    CountGroups = Table.RowCount(DistinctGroups)
in
    CountGroups)
```

Code 2: M code for distinguishing Single vs Cross-Group Incidents. Source: Author.

```
= Table.AddColumn(#"Added Custom", "cross/single_group", each
    if [DistinctGroupsCount] > 1 then "Cross Group"
    else "Single Group")
```

An addition to the table is the “major_incident” column, determined by the “major_incident_state” attribute (refer to Code 3). When this state is set to “Accepted”, the incident is classified as a major incident. Hence, the column indicates “Yes”, otherwise “No”.

Code 3: DAX formula for identifying major incidents. Source: Author

```
major_incident =
    IF(D_incident[major_incident_state] = "Accepted", "Yes", "No")
```

Dimension Table Incident Type

The “D_incident_type” table classifies incidents by their resolution scope. A ticket associated with multiple groups indicates a cross-group scenario. A parallel methodology was adopted to determine cross or single-supplier tickets based on the number of suppliers engaged in resolution. The table is manually created and integrated into the data model and links to the “D_incident” table through a one-to-many relationship based on “type_ID”.

5.1.2 Dimension Table Support Group

The Dimension Support Group table provides detailed attributes for all company support teams, including both supplier and internal teams. It features group names, types, active status, descriptions, and a hierarchy defined by parent group names. The hierarchy is crucial for determining each group's services provided. The “group_type” attribute identifies if a group is supplier-related and specifies the supplier.

In the existing data architecture, the support groups table retrieved from the database is supplemented by an additional table that lists specific services and suppliers linked to support groups. This table helps to correct any missing or incorrect hierarchy mappings within the ticketing tool. The data for this table is manually updated in an Excel file located on the company's SharePoint site. After accessing this file with the necessary company credentials, the information is brought into Power Query and then merged with the support group data by matching the group names. This step ensures that all group records are accurately represented.

To correctly classify supplier groups and determine service levels, the following columns were created in Power Query using M code:

Code 4: M code for adding a new column to classify supplier names. Source: Author.

```
= Table.AddColumn(#"Renamed Columns", "supplier_name", each
  if Text.Contains([group_type], "Supplier 1") and not Text.Contains([group_type], "HR Groups") then "Supplier 1"
  else if Text.Contains([group_type], "Supplier 2") then "Supplier 2"
  else if Text.Contains([group_type], "Supplier 3") then "Supplier 3"
  else if Text.Contains([group_type], "Supplier 4") then "Supplier 4"
  else "Other")
```

Code 5: M code to distinguish between supplier groups and other/internal groups. Source: Author.

```
= Table.AddColumn(#"Added Custom", "supplier_other", each
  if [supplier_name] = "Other" then "Other"
  else "Supplier")
```

When assigning services to groups, the following M code leverages a manual mapping table designed for specific groups, where services are explicitly outlined to facilitate precise classification. The approach ensures each group is tied to a service name, which enhances the ability to filter and analyse service levels across different suppliers. This structured method allows for dynamic updates, which is particularly useful when supplier group hierarchies in the ticketing tool require adjustment.

Code 6: M code for merging the “Support Groups” data with manually maintained group mappings. Source: Author.

```
= Table.NestedJoin(#"Added Custom1", {"name"}, manual_groups_mapping, {"support_groups"}, "manual_groups_mapping", JoinKind.LeftOuter)
```

Code 7: M code to map services to groups based on a set of criteria. Source: Author.

```
= Table.AddColumn(#"Expanded manual_groups_mapping", "service", each
  let
      supplier_name = [supplier_name],
      group_name = [name],
      parent_group_1 = [parent_group_1],
      parent_group_3 = [parent_group_3],
      parent_group_4 = [parent_group_4],
      manual_service = [service]
  in
      if manual_service <> null then manual_service
```

```
    else if supplier_name = "Supplier 2" and
      (Text.Start(group_name, 4) = "xxx " or
       Text.Start(group_name, 8) = "Support ") then "End User Services"
    else if supplier_name = "Supplier 2" and
            parent_group_1 = "yyy" then "Security Services"
    else if supplier_name = "Supplier 4" and
            group_name = "Group 123" then "Network Services"
    else if supplier_name = "Supplier 1" and (parent_group_3 = "Application" or
            parent_group_4 = "Application") then "Application Services"
    else if supplier_name = "Supplier 3" and (parent_group_3 = "Application" or
            parent_group_4 = "Application") then "Application Services"
    else "Other")
```

Dimension Tables Service and Supplier

After generating new columns within the “D_support_group” table that classify service and supplier information, distinct service and supplier names were extracted to form dedicated dimension tables, namely “D_service” and “D_supplier”. These tables were then assigned unique indexes, creating primary keys (“service_ID” and “supplier_ID”) that serve as stable, concise identifiers for each entry. Subsequently, these keys were integrated back into the support groups table, replacing verbose textual descriptions with numerical identifiers.

After importing the tables into the data model, one-to-many (1:M) relationships were established. This arrangement, with dimension tables branching off from a central fact table, is indicative of a snowflake schema.

5.1.3 Fact Table Incident Resolution

This table is fundamentally structured around Underpinning Contracts (UCs) related to Incident resolution. Each UC is initiated at key points, such as when an incident is opened and assigned to a support group. Within this table, each row represents a UC, detailing its stage, priority, corresponding incident number and incident state. It captures whether the UC has met its target resolution timeframe as well as records the support groups associated with the UC.

A distinct feature of this table is how it captures the evolution of an incident. For instance, if an incident undergoes a priority change, this is reflected in a new row with the same incident number and group but associated with a different UC. This is because UCs incorporate the priority level into their naming convention. Similarly, it captures the group changes of the incident. In essence, the table provides a comprehensive timeline of each incident, mapping out all the changes in priority, group, and state through different UCs. This design allows for a granular analysis of incident management, giving insights into how incidents evolve and are resolved.

The following SQL script was utilized to retrieve all records from the “incident_resolution” table for each incident number that has at least one entry with the incident state 'Closed' and “uc_start_time” on or after January 1, 2023.

Code 8: SQL script for extracting relevant incident resolution data. Source: Author.

```
SELECT *
FROM incident_resolution
WHERE incident_number IN (
    SELECT incident_number
    FROM incident_resolution
    WHERE incident_state = 'Closed' AND uc_start_time >= '2023-01-01'
)
```

Upon integrating the data into the model and completing common steps, several columns—"group_ID”, “service”, and “service_level”—were added using DAX expressions. The Incident Resolution table is linked to the Support Group table via a many-to-one relationship. Utilizing this linkage, the “group_ID” and “service” fields were brought into the Incident Resolution table with RELATED function (refer to Code 5).

Code 9: DAX formula for retrieving “group_ID” from dimension Support Group. Source: Author.

```
group_ID = RELATED(D_support_group[group_ID])
```

Code 10: DAX formula for retrieving “service” from dimension Service. Source: Author.

```
service = RELATED(D_service[service])
```

The attributes “service” and “priority” serve as a basis for determining the service level associated with each ticket. To transform these fields into a format suitable for visualizations, a DAX formula is employed to concatenate the service level and priority into a single, descriptive field. This newly calculated column facilitates a more intuitive display of data within dashboard visualizations, providing clear insights into incident service levels.

Code 11: DAX formula for constructing the service level names. Source: Author.

```
service_level =
VAR PriorityName =
    SWITCH(
        'F_incident_resolution'[priority],
        "P1", "Critical",
        "P2", "High",
        "P3", "Moderate",
        "P4", "Low"
    )
RETURN
 PriorityName & " Priority Incidents " & 'F_incident_resolution'[service]
```

5.1.4 Fact Table Incident Response

The “Incident Response” table contains the data regarding response times for Underpinning Contracts (UCs). Unlike the “Incident Resolution” table, it does not track the groups involved but rather focuses on capturing service, supplier, and priority as specified in the UC's naming structure. This information is extracted into separate columns through text parsing in Power Query. The table also logs the start and end times for each UC.

Based on the user requirements, for analytical purposes it is necessary to consider only the first response time associated with different services. As incidents may be reassigned and addressed by different groups, each will generate a new response time entry and a new UC. To identify the earliest start time for each incident, a “Group By” operation with a “Min” function is applied in Power Query. Once the data is imported into the model, service levels are delineated using a DAX formula similar to the one presented in Code 7, ensuring the focus remains on the first service group's response to each UC.

5.1.5 Dimension Table Date

The Date dimension is created in the data model using a DAX formula that merges unique end times from the “F_incident_resolution” and “F_incident_response” tables. This unified table, created via the UNION and SELECTCOLUMNS functions, ensures consistent date filtering across the report's visuals. A relationship of one-to-many (1:N) with the tables allows for comprehensive time-based analysis, as each date in the dimension can correspond to numerous incident records.

Code 12: DAX formula to form the Date table. Source: Author.

```
D_Date =
  DISTINCT(
    UNION(
      SELECTCOLUMNS(F_incident_resolution, "date", F_incident_resolution[end_time]),
      SELECTCOLUMNS(F_incident_response, "date", F_incident_response[end_time])
  ))
```

An additional column, “is_current_year”, identifies dates within the current year and serves as a discreet filter to restrict report views to the most recent data. If the result is '1', it signifies the date falls in the current year. This facilitates temporal analysis within the current operational year, enhancing the data model's utility for ongoing incident evaluation.

Code 13: DAX formula to determine current year status. Source: Author.

```
is_current_year =
  IF(
    YEAR([Date]) = YEAR(TODAY()), 1, 0
  )
```

5.1.6 Performance Metrics

For the incident resolution and response times within the service levels, minimum and expected targets were established, drawing from the benchmarks illustrated in Table 1 of Section 3.3. Given that the expected and minimum targets for resolution were consistent across services, new columns were populated in the Incident Resolution table with values of 96% and 94% respectively. On the contrary, the targets for response times necessitated more nuanced calculations to reflect varying service levels. The following DAX formula was constructed to assign expected target percentages based on the service level category. A similar approach was adopted to calculate the minimum targets, employing different percentage values as per the service level definitions.

Code 14: DAX formula for assigning expected target percentages. Source: Author.

```
expected_target =
VAR ServiceLevel = F_incident_response[service_level]

VAR IsAppServices = CONTAINSSTRING(ServiceLevel, "Application Services")
VAR IsCriticalOrHigh = CONTAINSSTRING(ServiceLevel, "Critical") ||
                       CONTAINSSTRING(ServiceLevel, "High")
VAR IsLowOrModerate = CONTAINSSTRING(ServiceLevel, "Low") ||
                      CONTAINSSTRING(ServiceLevel, "Moderate")

RETURN
IF(IsAppServices, 0.99,
   IF(IsCriticalOrHigh, 0.98,
      IF(IsLowOrModerate, 0.97,
         BLANK()
)))
```

To evaluate the performance against service level agreements, a metric was created to quantify incidents where the Underpinning Contract (UC) targets were not met. The metric, defined as “Failed UC Incidents”, counts incidents with a “Closed” state and a “Completed” UC stage that have breached the target duration. An additional ISBLANK function ensures that incidents without any data are accounted for as zeros rather than creating calculation errors.

Code 15: Metric showing the number of incidents that failed to meet the UC target. Source: Author.

```
rn_failed_incidents =
VAR FailedTickets = CALCULATE(
    DISTINCTCOUNT('F_incident_resolution'[incident_number]),
    'F_incident_resolution'[incident_state] = "Closed",
    'F_incident_resolution'[uc_stage] = "Completed",
    'F_incident_resolution'[has_failed] = "True"
)
RETURN
```

```
IF(ISBLANK(FailedTickets), 0, FailedTickets)
```

Code 16: Metric showing the percentage of incidents that failed to meet the UC target. Source: Author.

```
rn_failed_incidents_% =
DIVIDE(
    [rn_failed_incidents], DISTINCTCOUNT('F_incident_resolution'[incident_number]), 0
)
```

Conversely, to calculate the success rate of incident resolution, the "Successful Incidents Percentage" considers only those incidents that have been closed and the UC stage is completed without failing the UC targets. Similar metrics were developed to calculate the success rate and the number of incidents with failed UCs for incident response.

Code 17: Metric showing the percentage of incidents that successfully met the UC target. Source: Author.

```
rn_success_% =
VAR TotalTickets = DISTINCTCOUNT('F_incident_resolution'[incident_number])
VAR SuccessTickets = CALCULATE(
    DISTINCTCOUNT('F_incident_resolution'[incident_number]),
    'F_incident_resolution'[incident_state] = "Closed",
    'F_incident_resolution'[uc_stage] = "Completed",
    'F_incident_resolution'[has_failed] = "False"
)
RETURN
IF(TotalTickets = 0, 0, DIVIDE(SuccessTickets, TotalTickets, 0))
```

5.2 Dashboard Visualization

While Power BI distinguishes between "reports" and "dashboards", as delineated in Section 2.4.1, they are frequently treated interchangeably in this study. Drawing from the literature review on effective dashboard design, the layout has been crafted to align with the intended audience's needs and operational contexts. Clarity, simplicity, and user-friendliness are prioritized, along with the careful selection of visuals and consistency across the dashboard.

In the dashboard's design, color consistency is maintained through the use of a harmonized palette, with specific colors chosen for their clarity and distinction. The following colors, along with their varying opacity levels, have been strategically applied to ensure visual coherence across the dashboard's multiple pages: #D9B300, #577C79, #E66C37, and #666666.

It is important to highlight that, following the visualization process, the dashboard was subsequently published as an app in Power BI Service. In this format, the titles of the pages are presented as distinct sections within the app interface rather than on the pages themselves. This enhances the user experience by providing a clear and organized structure. The supplier performance assessment within the Power BI Desktop dashboard is detailed across four main pages, which include:

- Incident Overview
- Service Levels - Incident Resolution
- Service Levels - Incident Response
- Priority Changes

The main first page, depicted in Figure 4, serves as the central platform for analysing supplier-related incident resolutions. This interactive page, while appearing straightforward at first glance, uses filters to unlock a depth of information about incidents, supplier groups, and performance metrics.

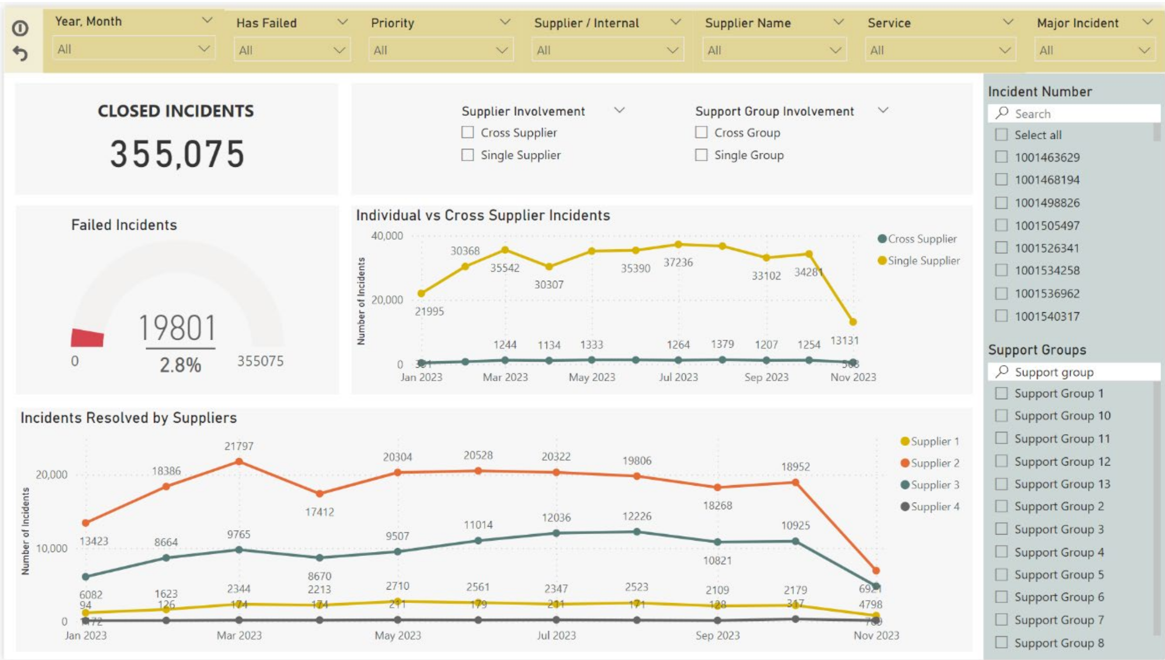


Figure 4: Dashboard's main performance assessment page — incident resolution overview.
Source: Author.

Positioned prominently on the top left is the total count of closed incidents. Directly beneath, a gauge visual depicts the ratio of failed incidents, offering both a numerical percentage and a graphic illustration utilizing metrics established in the preceding chapter. The line chart at the top contrasts individual and cross-supplier incidents over time, integrating data from the "Incident Type" and "Date" dimensions with the "Incident Resolution" fact table. The lower line graph presents a detailed tally of incidents resolved by suppliers, with distinct lines representing different suppliers.

Filterable through slicers at the top, as well as "Incident Number" and "Support Groups" on the right, the dashboard permits detailed scrutiny of supplier performance. Attributes like date, priority, service, supplier and group involvements enrich the analysis.

In the top left side, users can utilize the “Has Failed” filter to exclusively analyse incidents where the Resolution Underpinning Contracts (UCs) were not met (Figure 5). Furthermore, for an in-depth examination of a particular incident, the search functionality on the right side allows for entering a specific incident number. The dashboard will then dynamically filter to display all support groups that were involved in resolving that particular incident. Additionally, the “Major Incident” filter enables a focused view of only those incidents that have been escalated to a major incident status, thereby streamlining the assessment of critical issues.

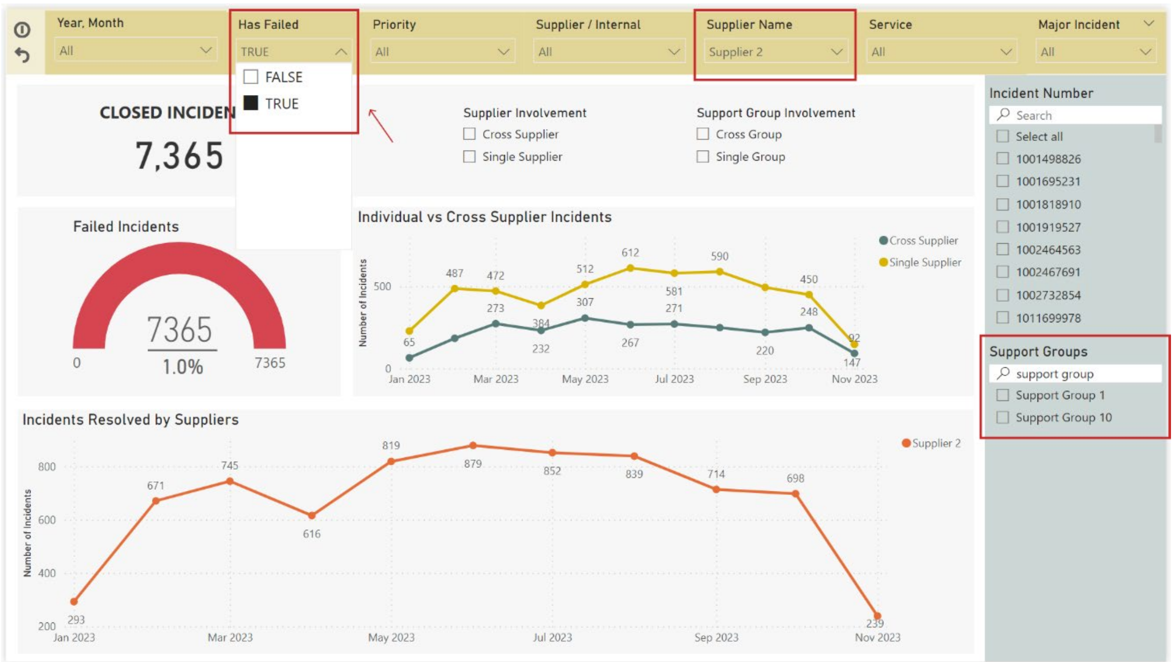


Figure 5: Dashboard’s main page filtered by “Supplier Name” and “Has Failed” slicers.
Source: Author.

At the top right corner of the dashboard, two interactive buttons enhance user navigation. The “Info” button, upon hover, informs users that holding the CTRL key allows for multi-selection within slicers — a functionality that may not be immediately apparent. It also provides brief explanatory notes on UC target specifics. The second button serves as a reset; when clicked, it clears all active filters, returning the dashboard to its original, unfiltered state.

The subsequent pages of the dashboard delve into the service-level performance of suppliers concerning incident resolution and response. These pages, displayed by tabular visuals, resonate with the format of Table 1 described in Chapter 3 of the thesis. This visualization approach was deliberately chosen for the dashboard to ensure user familiarity, as it mirrors the reporting format that users are already accustomed to.

Month	January				
Resolution Time - Service Levels	Expected Target	Minimum Target	Total Tickets	Failed Tickets	%
Application Services					
Critical Priority Incidents Application Services	96%	94%	3	0	100.00%
High Priority Incidents Application Services	96%	94%	401	26	93.52%
Medium Priority Incidents Application Services	96%	94%	1246	11	99.12%
Low Priority Incidents Application Services	96%	94%	3301	55	98.33%
End User Services					
High Priority Incidents End User Services	96%	94%	9	0	100.00%
Medium Priority Incidents End User Services	96%	94%	77	7	90.91%
Low Priority Incidents End User Services	96%	94%	9696	161	98.34%
Infrastructure Services					
Critical Priority Incidents Infrastructure Services	96%	94%	134	1	99.25%
High Priority Incidents Infrastructure Services	96%	94%	570	1	99.82%
Medium Priority Incidents Infrastructure Services	96%	94%	2274	1	99.96%
Low Priority Incidents Infrastructure Services	96%	94%	454	1	99.78%
Network Services					
Critical Priority Incidents Network Services	96%	94%	2	0	100.00%
High Priority Incidents Network Services	96%	94%	15	0	100.00%
Medium Priority Incidents Network Services	96%	94%	70	0	100.00%
Low Priority Incidents Network Services	96%	94%	29	1	96.55%
Security Services					
High Priority Incidents Security Services	96%	94%	3	0	100.00%
Medium Priority Incidents Security Services	96%	94%	2	0	100.00%

Total Incidents	Failed Incidents
1001476997	1021478687
1001485385	1061489430
1001491652	1141548015
1001522187	1171479224
1001527120	1271527436
1001540476	1391469579
1001558477	1401484602
1001561517	1431510328
1001572909	1441543175

Figure 6: Incident resolution Service Levels for individual supplier tickets (2nd page).
Source: Author.

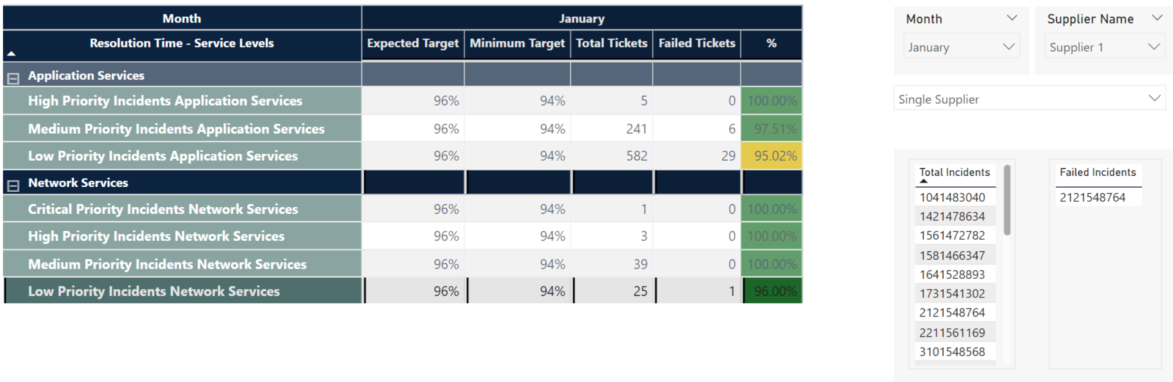
The table visual in Figure 6 delineates service levels by row, categorized under respective services. It displays expected and minimum target percentages for each service level alongside the count of total and failed tickets, which indicate incidents where the resolution of the ticket did not meet the UC target duration. The percentage column reflects the success rate, showing the calculations detailed in the previous section (5.1.6). Conditional formatting is applied to this column to visually represent whether service levels met, exceeded, or failed to reach the expected and minimum targets: green for exceeded or met, yellow for between targets, and red for below the minimum target. Similarly, this is done on the Incident Response page.

Code 18: Metric for conditional formatting logic in Incident Resolution page. Source: Author.

```
rp_color_code =
VAR SuccessRate = [rn_success_%]
VAR ExpectedTarget = MAX('F_incident_resolution'[expected_target])
VAR MinimumTarget = MAX('F_incident_resolution'[minimum_target])

RETURN
IF(SuccessRate >= ExpectedTarget, 1,
    IF(SuccessRate >= MinimumTarget && SuccessRate < ExpectedTarget, 2,
        IF(SuccessRate < MinimumTarget, 3, BLANK()
)))
```

Adjacent to the table, filters for “Month”, “Supplier Name”, and “Single or Cross-Supplier” allow for detailed data slicing. As Figure 7 exhibits, when filtered for January, “Supplier 1” and “Single Supplier”, the data reflects tickets resolved solely by this supplier, indicating non-collaborative resolutions. The data also indicates the services that the supplier provides.



Month	January				
Resolution Time - Service Levels	Expected Target	Minimum Target	Total Tickets	Failed Tickets	%
Application Services					
High Priority Incidents Application Services	96%	94%	5	0	100.00%
Medium Priority Incidents Application Services	96%	94%	241	6	97.51%
Low Priority Incidents Application Services	96%	94%	582	29	95.02%
Network Services					
Critical Priority Incidents Network Services	96%	94%	1	0	100.00%
High Priority Incidents Network Services	96%	94%	3	0	100.00%
Medium Priority Incidents Network Services	96%	94%	39	0	100.00%
Low Priority Incidents Network Services	96%	94%	25	1	96.00%

Figure 7: Filtered view of incident resolution Service Levels by supplier and time. Source: Author.

A distinctive feature on these pages is the interactive display of total and failed incident tickets. Selecting a specific category, such as low-priority incidents in network services, highlights the corresponding aggregate counts, as well as the actual failed incident numbers in the adjacent visual. On the right, the tables for “Total Incidents” and “Failed Incidents” update in response to selections within the service level table, offering a quick reference to specific incidents related to the selected service level criteria.

Month	January				
Response Time - Service Levels	Expected Target	Minimum Target	Total Tickets	Failed Tickets	%
High Priority Incidents Messaging Services	98.00%	96.00%	44	0	100.00%
Low Priority Incidents Messaging Services	97.00%	94.00%	88	13	85.23%
Moderate Priority Incidents Messaging Services	97.00%	94.00%	10	0	100.00%
Monitoring Services					
High Priority Incidents Monitoring Services	98.00%	96.00%	14	1	92.86%
Low Priority Incidents Monitoring Services	97.00%	94.00%	1	0	100.00%
Moderate Priority Incidents Monitoring Services	97.00%	94.00%	3	0	100.00%
Security Services					
Critical Priority Incidents Security Services	98.00%	96.00%	1	0	100.00%
High Priority Incidents Security Services	98.00%	96.00%	84	10	88.10%
Low Priority Incidents Security Services	97.00%	94.00%	64	0	100.00%
Moderate Priority Incidents Security Services	97.00%	94.00%	2	0	100.00%
Server Services					
Critical Priority Incidents Server Services	98.00%	96.00%	199	6	96.98%
High Priority Incidents Server Services	98.00%	96.00%	619	45	95.15%
Low Priority Incidents Server Services	97.00%	94.00%	120	1	99.17%
Moderate Priority Incidents Server Services	97.00%	94.00%	1143	4	99.83%
Storage Services					
Critical Priority Incidents Storage Services	98.00%	96.00%	46	3	93.48%
High Priority Incidents Storage Services	98.00%	96.00%	60	0	100.00%
Low Priority Incidents Storage Services	97.00%	94.00%	11	1	90.91%
Moderate Priority Incidents Storage Services	97.00%	94.00%	1195	53	96.40%

Figure 8: Service Levels for incident response times (3rd page). Source: Author.

The fourth page of the dashboard is specifically tailored to examine priority alterations during the incident resolution process. It addresses how changes in incident priority, such as a modification from moderate to low, affect the target duration for resolution. These adjustments are critical as they can extend the time allocated for suppliers to address incidents, thereby influencing the adherence to the target durations in Underpinning Contracts (UCs). The text on the top left of Figure 9 provides examples of standard target durations by priority level, offering a reference point for what constitutes a timely resolution.

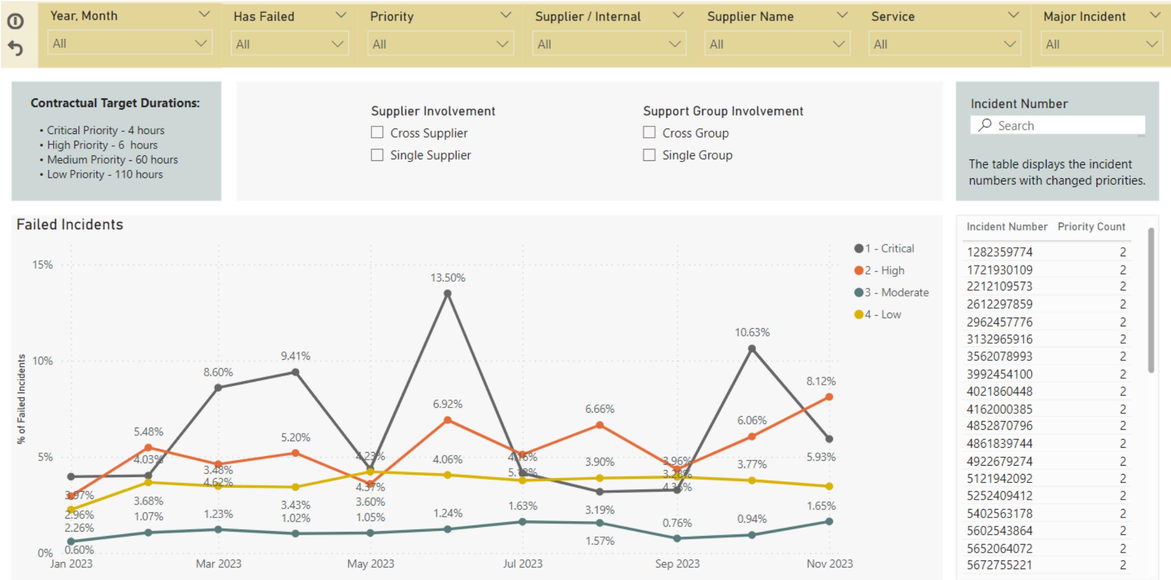


Figure 9: Analysis of priority changes and their impact on incident resolution (4th page). Source: Author.

Central to this page is a line chart visualizing the percentage of failed incidents across different priority categories. Such visualization facilitates an immediate grasp of how priorities impact the success rate of meeting UC targets. In the top right corner, a dedicated section lists incidents that have undergone priority changes. This feature allows users to investigate specific incidents that have been reclassified and assess whether these changes correlate with an increase or decrease in the incident failures to meet UC targets.

Highlighting priority changes provides insights into the dynamics of incident management, revealing whether adjustments in priority are used strategically to manage workload and expectations. It also informs discussions on supplier performance, differentiating between failures due to inherent service issues and administrative re-prioritization. Moreover, this can help identify patterns or trends that might indicate systemic issues in incident categorization or potential areas for process improvement.

5.3 Data Protection and Management

The source data for the dashboard is secured at its origin, with strict access controls in place both in the database and on the SharePoint site, ensuring that the information is accessible only to authorized users.

Before publication, the dashboard also underwent data protection measures, including the implementation of Row-Level Security (RLS). This security layer ensures data visibility is tailored to specific user groups. For instance, Supplier 1 teams can be restricted to viewing only data related to their service levels within the service level tables. This was achieved by defining roles within Power BI Desktop, which employed DAX expressions to filter the data accordingly. Upon publishing to the Power BI Service, these roles were then mapped to the respective supplier and internal distribution lists to maintain security and integrity.

The dashboard was initially released in a dedicated Development workspace within Power BI Service, accessible solely to the dashboard's future maintenance team. It then progressed to the Testing environment, where the supplier management team, along with select members from the supplier side, were granted access for evaluation purposes. This phase allows for critical feedback and any necessary modifications.

Following testing, the dashboard is transferred to the Production workspace. Here, it is published as an application accessible to all IT staff and associated suppliers. The visibility of specific dashboard pages is controlled through app audiences defined during publication in Power BI Service, ensuring that each user group encounters only the data and pages relevant to their role. This segmented approach to information distribution ensures both the utility of the dashboard for diverse user groups and the security of sensitive data.

Upon publishing the dashboard to the Power BI Service, the data refresh schedules were configured in the settings for daily updates. This functionality was enabled by choosing the correct connection types during the setup, which is essential for activating the automatic refresh feature. Given that the dashboard is utilized by users across various global locations, the refresh schedule was strategically set for 6 a.m. Prague time. This timing was chosen to ensure that users, regardless of their time zone, would have access to the most updated insights during their respective business days.

Moreover, with the introduction of the “Current Year” filter in Service Level pages, the dashboard is set to display only the relevant year's data. Hence, in the upcoming year, it will automatically update to show next year's data, ensuring continuous relevance without the need for manual adjustments. This setup guarantees that the dashboard remains up-to-date and functional at all times, providing users with the latest data from our sources.

The data model in Power BI has been designed with flexibility and scalability in mind, allowing for easy expansion to encompass additional service levels and integration with various ITSM processes in the future.

5.4 Technical Documentation

Throughout the development of the supplier performance assessment dashboard for this thesis, comprehensive technical documentation was created for the company and the team responsible for its maintenance. This documentation methodically outlines all critical aspects, such as the process for acquiring required data to construct the dashboard, procedures for obtaining necessary accesses, an inventory of currently available data, and other important considerations.

A significant section of the technical documentation is dedicated to a detailed description of the data transformation steps conducted within Power Query. This guidance is particularly vital for any future incorporation of new service levels and adjustments to the data model. Furthermore, the documentation includes a description of the metrics, columns and tables created post data load using DAX in Power BI Desktop, as well as the relationships between these tables.

The documentation further details the process of publishing the dashboard to the Power BI Service, outlining the security protocols employed and the configuration of data refresh schedules. This manual is designed to facilitate ongoing dashboard management and enhancements, ensuring the tool remains aligned with the evolving analytical requirements of the company.

6 Dashboard Assessment

This chapter delves into the assessment of the dashboard from the perspective of its key users. The aim is to understand user interaction with the new tool and gather actionable feedback for ongoing improvements. It is noteworthy to mention that feedback was consistently collected throughout the entire creation process of the dashboard. However, the emphasis of this chapter is primarily on the feedback and insights obtained after its deployment.

6.1 Deployment and Testing

The dashboard was deployed to the production environment and actively used by the key stakeholders for a trial period of one month. During this phase all four suppliers were asked to report on their service levels using the new dashboard, with a focus on comparing the new method of data representation and analysis to their previous practices. Concurrently, some internal employees were assessing services related to their roles, contrasting their findings with the ticketing tool to identify discrepancies or areas for improvement. This practical, user-led testing phase was critical, serving as a verification process to confirm the accuracy of the dashboard data against the ticketing system. The trial period allowed users to integrate the dashboard into their regular reporting routines and provided a sufficient timeframe to evaluate its performance.

Testing scenarios were based on actual performance reporting use cases. This approach allowed for the observation of user interactions with the dashboard, identifying both its strengths and areas requiring refinement. Scenarios included, but were not limited to:

- **Resolution and Response Time Analysis**: Suppliers were asked to use the appropriate filters on the dashboard's main page, determine their monthly resolved incident counts across different service levels and compare these figures against the set targets and data presented on the service level pages of the dashboard.
- **Supplier Performance Tracking**: Users tracked the performance of their support team over the past quarter, analyzing metrics like the number of closed incidents and the proportion of failed incidents. They were also asked to compare incidents resolved solely by their group with those involving collaboration with other groups.
- **Priority Change Impact**: Users examined the failed incidents that underwent priority changes with the aim of evaluating the impact these changes had on resolution times and adherence to UC target durations.
- **Major Incident Analysis:** Users were asked to analyze major incidents to validate the dashboard's data against activities logged in the ticketing tool, including resolutions carried out by various support groups.

The completion of these scenarios by users was essential before they participated in the feedback survey. Engaging with the dashboard through these tasks ensured that users had substantial hands-on experience, which provided a solid base for them to offer feedback.

6.1.1 Feedback Collection

Subsequent to the trial period, user feedback was gathered through a structured mixed-method survey. The first part of the survey consisted of statements rated on a Likert scale from 1 (strongly disagree) to 5 (strongly agree), allowing users to evaluate their experience quantitatively on various aspects of the dashboard, such as ease of navigation, clarity of information, and overall usefulness. This quantitative part encompassed questions 1 through 8, as outlined in Table 4.

The latter part of the survey featured open-ended questions conducted via meetings and email correspondences. This qualitative engagement aimed to gather detailed insights into user preferences, challenges encountered, and additional features they might find beneficial. Approximately 50 key users, including both suppliers and internal staff from different teams and divisions, actively engaged in the survey.

Table 4: Set of survey questions to gather user feedback. Source: Author.

N	Question
1	The dashboard navigation is intuitive and user-friendly.
2	The information presented on the dashboard is accurate, clear and understandable.
3	The dashboard has a visually appealing layout and design. The color codes and conditional formatting are helpful and accurate in highlighting critical data points.
4	The dashboard provides all the necessary information for support group performance analysis.
5	The dashboard effectively facilitates the assessment of incident resolution and response times.
6	The performance targets and metrics are presented in a way that supports my decision-making process.
7	The filtering and search features provided in the dashboard are appropriately selected and significantly enhance my analytical capabilities.
8	The dashboard makes it easy to identify trends or issues with supplier performance.
9	What additional information/functionalities would you like to see included in the dashboard?
10	Did you encounter any challenges/uncertainties while using the dashboard?
11	How often do you use the dashboard, and does it meet your expectations in terms of data refresh and accuracy?
12	How would you rate the overall usefulness of the tool in managing supplier performance?
13	Would you recommend any changes to improve the dashboard's user experience?

6.2 Analysis of User Feedback and Insights

Analysis was conducted on the responses from key users, and below in Table 5 are presented the average scores and the most frequent (mode) responses for each question. The results indicate that while average scores are generally high, the most frequent response to each question may vary.

Table 5: Summary of feedback survey results. Source: Author.

N	Question Description	Average Score	Most Frequent Response
1	Ease of Dashboard Navigation	4.14	4
2	Clarity of Information Presented	3.98	4
3	Visual Appeal and Layout: Color Codes and Conditional Formatting	4.30	4
4	Provision of Necessary Information	4.10	3
5	Effectiveness in Assessing Incident Resolution and Response Times	4.90	5
6	Presentation of Performance Targets and Metrics	4.20	4
7	Utility of Filtering and Search Functionalities	4.66	5
8	Facilitation of Identifying Trends/Issues with Supplier Performance	4.23	4

The average scores from the questions indicate a high level of satisfaction with the dashboard. Particularly noteworthy are:

- Effectiveness in Assessing Incident Resolution and Response Times (4.90): Users found the dashboard exceptionally effective for this purpose, indicating its strong alignment with the core objectives of the dashboard.
- Utility of Filtering and Search Functionalities (4.66): This high score suggests that users find these features not only user-friendly but also integral to their analysis, enhancing their ability to navigate and interpret data.
- Visual Appeal and Layout (4.30): The aesthetically pleasing and well-organized interface is clearly appreciated by the users, which can be crucial for user engagement and long-term adoption. The positive response also suggests that the selection of graphs and visuals effectively guides users to essential data.

Question 9: In response to the comparatively lower score for the provision of necessary information (4.10), users specifically addressed their needs in question 9. They expressed a desire for the dashboard to include more comprehensive IT service management data, especially focusing on service request ticket analysis and subsequent integration of problem and change service levels, similar to the current service level tables. However, users themselves acknowledged that expanding the dashboard to encompass these additional service levels is of lower priority and can be addressed in future updates. This suggests that the limitation is more temporal, with plans to incorporate these elements in subsequent phases of development.

Furthermore, there was a request for incorporating notes about the logic behind support group mappings directly within the dashboard for better understanding and future structural enhancements. This topic was thoroughly discussed in a meeting, leading to the decision that additional notes within the dashboard were unnecessary, as the underlying logic and methodology were adequately clarified during the discussion. This resolution demonstrates active engagement with user feedback and a commitment to ensuring the dashboard remains user-friendly and self-explanatory.

Another suggestion was the ability to pinpoint the specific team responsible for failures in cross-supplier tickets. While the dashboard currently identifies all parties involved in such tickets, the integration of data that specifies which exact team failed is currently unfeasible due to limitations in data resources.

Question 10: Addressing question 10, which focused on any difficulties or confusion experienced while using the dashboard, the primary issue identified was related to the filters for cross/individual supplier and group involvement. It was found that the original naming of these filters had not been clearly explained to all supplier groups, leading to some confusion. This observation is reflected in the relatively lower average score of 3.98 for question 2. To resolve this, the filter names were updated to "Supplier Involvement" and "Support Group Involvement", respectively, to enhance clarity and user comprehension.

Question 11: For question 11, which inquired about the frequency of dashboard usage and its satisfaction in terms of data refresh and accuracy, the predominant response indicated usage twice or once weekly. This frequency varied with specific periods of the month, particularly during weeks with heightened reporting activity, where usage increased. Users confirmed the accuracy of the data as they cross-verified it with the ticketing tool. The dashboard's current refresh rate on a daily basis was positively noted for its timeliness, aligning not only with decision-making needs but also serving effectively for reporting purposes.

Question 12: Regarding question 12, which explored the dashboard's overall usefulness in assessing supplier performance, feedback particularly emphasized its effectiveness as a tool for automated reporting. This perspective was strongly voiced by internal employees who expressed confidence in the dashboard's data reliability and appreciated its enhanced visibility features.

Users noted the dashboard's significant value in incident management analysis, highlighting its capability to display all involved support groups. Compared to the dashboard, this functionality is not available in the existing ticketing tool. This feature was seen as particularly advantageous, enhancing the comprehensiveness of the analysis. The dashboard was recognized as a crucial resource for suppliers to report on and be evaluated against their performance. However, one of the common improvements mentioned in the responses to this question was the inclusion of other service level deliverables in a similar tabulated format to facilitate a complete assessment of overall performance. This feedback indicates a desire for the dashboard to evolve into an even more integrated tool that covers a broader spectrum of service management aspects.

Question 13: In terms of potential improvements to enhance the dashboard's user experience, the feedback indicated a high level of satisfaction with its current user friendliness. Several factors may contribute to this positive feedback. Firstly, the widespread familiarity with Power BI among the users likely contributed to the ease of navigating and understanding the dashboard. This pre-existing knowledge of the tool's interface and functionalities made the transition to using the new dashboard more seamless. Secondly, the design of the service level pages (the second and third pages of the dashboard) closely resembles the actual service level reports traditionally submitted by suppliers. This consistency in presentation style meant that users were already comfortable with the format and could easily interpret the data.

Lastly, the positive user experience might also stem from the inclusive approach taken during the dashboard's development. From the outset, the development process involved continuous engagement and feedback from the users. This collaborative strategy ensured that the final product was tailored to meet user requirements and preferences, enhancing its overall usability.

6.3 Evaluation and Fulfillment of Dashboard Requirements

This section assesses the dashboard's success in meeting the set objectives and outlines how it overcomes the challenges of the previous reporting methods. Additionally, it outlines the roadmap for addressing the unmet user requirements.

The initial approach of reporting and assessing supplier performance within the company faced numerous challenges and limitations, as detailed in Section 3.6. It is heavily manual, predominantly relying on reports extracted from the company's ticketing tool and analyzed in Excel. This process involved manually filtering incident tickets, categorizing them, and then interpreting these data points to evaluate supplier performance. The method was hindered by limitations such as the ticketing tool's inability to display all involved groups in a resolution, a lack of standardization across different suppliers, restrictive data extraction limits, and inadequate visualization capabilities. Additionally, the reliance on static tools and manual data handling raised concerns about data reliability and resulted in time-consuming processes, underscoring the need for a more efficient and automated process.

The proposed dashboard represents a significant improvement over the initial approach. It automates the reporting process, eliminating the manual effort previously required from suppliers. This automation is complemented by enhanced visibility, providing stakeholders with comprehensive insights into supplier performance. The dashboard's design facilitates the integration of all necessary data, ensuring reliability and up-to-date information.

The primary objective of this thesis was to design and implement a Self-Service Business Intelligence (SSBI) solution for analyzing IT service suppliers' performance. Key goals included automating the reporting process and improving visibility for the supplier management team and stakeholders.

Table 6 combines the user requirements initially presented in Tables 2 and 3 of Chapter 4. It displays these requirements in a concise format, encompassing both business, data, and functional needs. The table not only summarizes these user stories but also provides an update on their fulfillment status based on user feedback. Key users have confirmed that all high and medium priority requirements have been satisfactorily met. Based on their feedback, a low-priority requirement concerning the differentiation of individual and cross-group tickets has also been met. This was achieved through a similar approach used for identifying cross-supplier tickets. However, two low-priority requirements remain unfulfilled due to data unavailability and time constraints. Nevertheless, the dashboard's foundation allows for future expansions to address these areas.

Table 6: Overview of fulfillment status for dashboard requirements. Source: Author.

Requirement	**Priority**	**Status**
Reliable reporting tool for independent supplier performance assessment	High	Met
Interactive dashboard with comprehensive filtering capabilities	High	Met
Monthly incident closure tracking	High	Met
Analysis of incidents resolved or failed by each supplier	High	Met
Specific service level analysis for suppliers	High	Met
Resolution tracking for each priority level	High	Met
Comparison of single vs cross-supplier ticket resolution	High	Met
Analysis and visibility of support groups involved in ticket resolution	High	Met
Incident response time calculation based on first resolver group	High	Met
Daily dashboard refreshes for up-to-date metrics	High	Met
Full automation of reporting	High	Met
Security-constrained views for specific supplier data	High	Met
Separate tracking for major incidents	Medium	Met
Analysis of tickets with priority changes	Medium	Met
Understanding collaborative dynamics in ticket resolution (support groups)	Low	Met
Display of all service management metrics (request, problem, change)	Low	Unmet
Identification of user dissatisfaction from supplier resolutions	Low	Unmet

The first unmet requirement, relating to the display of comprehensive service levels, could not be addressed within the given timeframe. Incorporating additional IT service management processes like service requests, problems, and change management necessitated a more detailed analysis to thoroughly understand and accurately represent all the metrics. Nevertheless, the dashboard's data model was designed for scalability, allowing for the future integration of these elements. The existing data sources are comprehensive enough to support this expansion, and the detailed technical documentation prepared will facilitate this enhancement.

The second unmet requirement, involving the integration of customer satisfaction metrics, was delayed by the unavailability of relevant data in the database. A proposed two-month extension can resolve this, enabling the addition of customer satisfaction metrics as another service level. The company's data request process must be navigated to make this data available for dashboard integration.

Considering the objectives set out at the beginning of this thesis, it is clear that the work has accomplished its goals. All high and medium priority requirements have been met, representing a significant progression from the original manual reporting processes. The dashboard not only delivers on the promise of automated reporting but also provides the enhanced visibility that was initially aimed for, fulfilling the goals of developing an effective SSBI solution for supplier performance analysis.

7 Recommendations

This chapter outlines future directions for the development and utilization of the SSBI solution. Based on the user requirements and feedback, it is suggested that the scope of the dashboard be expanded to include additional IT service management processes such as service requests, problems, and change management. This expansion would provide a more comprehensive view of supplier performance. The existing data model within Power BI has been designed to facilitate such expansion, ensuring a seamless integration of these new processes.

Furthermore, it is proposed to incorporate customer satisfaction metrics into the dashboard. Each time a ticket is resolved, customers are invited to rate their satisfaction through a single-question feedback mechanism. By integrating these ratings, the dashboard would not only reflect quantitative data but also provide valuable insights into the qualitative aspects of how users perceive the resolution of their tickets. This enhancement will enrich the dashboard's capability in evaluating the overall quality of services provided by suppliers, giving a more complete picture of their performance.

An additional recommendation concerns the process of retrieving Service Level target percentages. Given that these targets are subject to annual reviews and potential updates, and considering the possibility of adding new service levels from other ITSM processes, it is advised to maintain a dedicated Excel document within the company's SharePoint for target updates. This approach would allow for automatic updates of the targets in the dashboard each year, thereby streamlining the process by eliminating the need for manual modifications in the DAX formulas.

Finally, it is vital to ensure continuous support and updating of the dashboard, especially as contracts with suppliers evolve and as the company's data and reporting requirements change. To maximize the effectiveness of the dashboard, ongoing user training sessions should be conducted, and updates in the technical documentation should be made available. This approach will assist users in fully exploiting the dashboard's capabilities for informed decision-making and analysis.

Conclusion

The primary goal of this thesis was to design and implement a Self-Service Business Intelligence (SSBI) solution for visualizing and analyzing IT service suppliers' performance in the company. This led to two main sub-objectives: automating the reporting process to reduce dependency on supplier-provided data and enhancing data visibility to meet specific stakeholder requirements. These objectives were achieved by addressing high and medium priority needs identified through discussions with key users and reaching full automation in the reporting process. The SSBI solution is now integrated and used by suppliers and internal employees within the company. The dashboard compiles essential data, ensuring reliability and providing current information for all stakeholders.

To reach these goals, an extensive literature review was conducted on business intelligence, the implementation of SSBI applications, and Power BI, the tool selected for dashboard development. An analysis of the company's performance assessment methods and incident management preceded the analysis of user requirements. Through discussions with key users, specific requirements were gathered, leading to the dashboard's design and implementation. User testing and survey feedback helped evaluate the fulfillment of these requirements.

The primary outcome of this thesis is a fully automated Power BI dashboard published in the Power BI Service as an app. It meets the objectives and user requirements, offering numerous benefits over previous methods. The dashboard enhances operational efficiency by reducing human error and saving time. It provides a granular view into ticket categories, a feature not possible with the previous ticketing tool, thus offering detailed visibility into tickets resolved by both multiple and single suppliers. This level of insight extends to the specific support groups involved in each resolution and includes monitoring changes in ticket priorities, which is essential for maintaining contract compliance. The supplier management team now relies on these dashboards for accurate, up-to-date data, promoting a data-driven approach to supplier performance management. Furthermore, the dashboard's service level metrics are developed from established reporting methods, which reduces the need for additional user training. With a focus on security, this solution ensures sensitive information is only accessible to authorized personnel. This was achieved by leveraging Power BI's advanced features like Row-Level Security for customized data access and audience selection in Power BI Service. Additionally, technical documentation has been created to support the dashboard's ongoing development and future enhancements.

The agile-centric approach and iterative development cycles were key in aligning the dashboard with stakeholders' needs and improving upon initial reporting methods. User feedback has identified future enhancements, such as expanding the scope to cover other ITSM processes and integrating customer satisfaction ratings. These enhancements, among others discussed in this thesis, will offer a more comprehensive view of supplier performance.

The successful implementation of this project demonstrates the potential of SSBI applications in transforming data analytics and decision-making processes in corporate environments. This work contributes to the organization's operational efficiency and the broader academic understanding of applying Business Intelligence in practical, real-world scenarios.

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Appendices

Appendix A: M Code Snippets for Power Query

This appendix presents the M code snippets used in Power Query for various data manipulation tasks. Code 19 calculates the number of unique suppliers involved in each incident, which is a key feature in assessing supplier involvement.

Code 19: M code to count distinct suppliers associated with an incident. Source: Author.

```
= Table.AddColumn(#"Grouped Rows", "DistinctSuppliersCount", each let
    IncidentData = [GroupedData],
    DistinctSuppliers = Table.Distinct(IncidentData, {"supplier_ID"}),
    CountSuppliers = Table.RowCount(DistinctSuppliers)
in
    CountSuppliers)
```

Below code differentiates incidents managed by a single supplier from those that involve multiple suppliers, providing insights into the complexity of incident resolution.

Code 20: M code for distinguishing Single vs Cross-Supplier Incidents. Source: Author.

```
= Table.AddColumn(#"Added Custom", "cross/single_supplier", each
    if [DistinctSuppliersCount] > 1 then "Cross Supplier"
    else "Single Supplier")
```

Appendix B: SQL Scripts for Data Retrieval

This appendix presents SQL scripts used to extract necessary data from the database for analysis and reporting. The SQL script in Code 21 retrieves closed incident data since the beginning of 2023.

Code 21: SQL script for extracting relevant incident data. Source: Author.

```
SELECT *
FROM incident
WHERE state = 'Closed' AND opened_at >= '2023-01-01'
```

This SQL script in Code 22 focuses on extracting data relevant to incident response times, essential for evaluating compliance with underpinning contracts.

Code 22: SQL script for extracting relevant incident response data. Source: Author.

```sql
SELECT *
FROM ticket_sla
WHERE type = 'Underpinning contract'
  AND collection = 'incident'
  AND start_time >= '2023-01-01'
  AND target = 'Response'
  AND stage = 'Completed'
```